

FIELD TECHNOLOGY V5

Volume 03 — Operator (Chapters 9–12)

Zachary Robert Geurts · Field Primer PDF edition

Stability Under Load — FCC & Tesla

CHAPTER 9 · STABILITY UNDER LOAD — FCC & TESLA

Learning objectives

- State wave and diffusion CFL inequalities and why the host enforces them before `vkCmdDispatch`.
- Map FCC floats to `data_bus[16–23]` and describe harmonics guard scaling.
- Quote Tesla constants and explain directional damping on spiderweb edges.
- Contrast PropalacticScale knob with planetary RF visual shell.
- Describe KILROY kernel FCC parallel and entropy feedback clamps.

IN THIS CHAPTER

- | | |
|---|--|
| 1. Introduction — stability is offense ethics | 8. PropalacticScale — honesty label |
| 2. Wave CFL — intuition on a grid | 9. Tesla valve — constants and publication |
| 3. Diffusion CFL — why Δx^2 matters | 10. Tesla on Flow fabric |
| 4. Harmonics guard implementation posture | 11. KILROY FCC — kernel parallel |
| 5. FCC floats — complete slot map | 12. Failure modes — NaN theology |
| 6. GateFidelity — sharp gates and stability | 13. Operator drills |
| 7. EntropyFloor — second law as engineering | 14. Chapter summary |

On the way — what you will learn

Before beauty runs hot, the host refuses NaN theology. On the way you will enforce CFL wave and thermo guards, publish Tesla valve bias to the bus, and parallel KILROY FCC discipline at the kernel boundary.



On the way — Tesla valve metaphor

- Courant-Friedrichs-Lewy harmonics guard
- TESLA_R_FORWARD / REVERSE metaphor
- data_bus slots 31 and 34
- Stability under load without censorship

Introduction — stability is offense ethics

Offense without stability is vandalism against your own field. Chapter 7 placed the CFL harmonics guard on the dispatch spine; Chapter 9 teaches the science, constants, and operator drills behind that guard. FCC here means **Field Control Constants** — analog floats packed to data_bus [16–23] — not Federal Communications Commission, though NEXUS ITU-R path loss docs live beside RF teaching in Chapter 6.

Before fabric evolution each dispatch_canvas(), host computes:

$$\text{waveCFL} = c \cdot \Delta t / \Delta x \leq 1 \cdot \text{thermoCFL} = \alpha \cdot \Delta t / \Delta x^2 \leq 1$$

Violations scale down unsafe parameters. Plain English: the engine refuses diffusion so hot the simulation explodes into NaN. This is numerical ethics — creditor debt to Courant, Friedrichs, and Lewy. CFL creditor page

Cross-links: Chapter 3 thermo knobs, Chapter 7 dispatch placement, Chapter 8 bus slots, Chapter 10 spiderweb Tesla edges, Chapter 12 honesty rows for Propalactic and Tesla metaphor.



Figure 9.1 — Directional damping metaphor with constants in `FieldRtxFieldAbs.hpp`.

Wave CFL — intuition on a grid

Explicit wave integrators allow information to travel at most one cell per step. If $c\Delta t$ exceeds Δx , the scheme tries to move wave energy farther than the mesh can represent — instability follows. Host reads render resolution for Δx proxy, `WaveSpeed` for c , `TimeScale` for effective Δt multiplier. Product $\text{waveCFL} \leq 1$ is the guard line.

Operators who crank `WaveSpeed` in prompt without reading `stderr` may never see NaN — host scales c down first. That is care, not censorship. Compare list `AnalogFields` after aggressive set — admitted value may differ from requested.

Diffusion CFL — why Δx^2 matters

Diffusion stiffens faster as meshes refine, $\text{thermoCFL} = \alpha\Delta t/\Delta x^2 \leq 1$ shrinks admissible Δt when resolution upscale halves Δx . `RayCanvas` adaptive 4K path interacts with guard — Chapter 7 warned adaptive scale is not vanity. Adept operators watch `STATUS` when changing resolution and `TimeScale` together.

Harmonics guard implementation posture

From OptionsMenu.hpp commentary: FCC floats are pre-conditioned in dispatch_canvas before GPU. Hard caps include waveSpeed $\in [0.01, 2.0]$, dT ≤ 0.033 , inject strength cooperation. Body-temperature seeding and similar simulation flavor remain labeled Metaphor in Chapter 3 — not hidden as SI measurement.

Input	Guard effect
TimeScale $\uparrow$	Effective Δt $\uparrow$ — both CFL products $\uparrow$
ThermoAlpha $\uparrow$	thermoCFL $\uparrow$
WaveSpeed $\uparrow$	waveCFL $\uparrow$
Resolution $\uparrow$	Δx $\downarrow$ — thermoCFL $\uparrow\uparrow$
InjectStrength $\uparrow$	May clamp with coupling workload

FCC floats — complete slot map

```
[16] TimeScale [17] ThermoAlpha [18] WaveSpeed [19] GateFidelity
[20] EntropyFloor [21] InjectStrength [22] PropalacticScale [23] FieldCoupling
```

Each float is mirrored for guest HUD and grep. Shader bindings 8–10 receive admitted values post-guard. FieldSocket path and Classic path share guard philosophy even when push layout differs.



GateFidelity — sharp gates and stability

0 = soft analog gates; 1 = transistor-sharp Flow gates. Sharpness changes gradient stories and indirect coupling workload — can raise effective field work and prevMaintCost. Stability is not only CFL; coupling spikes can challenge entropy proxy without violating CFL if clamps already admitted parameters.

EntropyFloor — second law as engineering

Minimum irreversible noise in fabric — Chapter 3 entropy floor. Prevents unphysical reversibility. Works with CFL as dual ethics: one limits step size, one enforces noise floor.

PropalacticScale — honesty label

Large-wavelength forcing on Phi. Chapter 12 honesty: dynamics knob, not cosmic oracle. Moves fabric; does not replace packet field or GPS precision maps. Metaphor when marketing says cosmic weave.

Tesla valve — constants and publication

```
TESLA_R_FORWARD = 0.18    TESLA_R_REVERSE = 3.2    FIELD_PHI_MILLI = 618
TESLA_R_FWD_MILLI = 180   TESLA_R_REV_MILLI = 3200
```

teslaBias() in updateHardwareFromAnalogFields dampens reverse flow on spiderweb edges more than forward. Published to data_bus[31] and [34]. TeslaBiasStrength FCC knob mirrors coherence.

Fluidic diode metaphor — not literal chassis part.

Tesla creditor

Tesla on Flow fabric

Flow channel mixes gradient magnitude with GateFidelity and Tesla relaxation — Chapter 3 per-texel evolution. Directional preference appears in color stories operators see on Classic canvas and in edge util mirror Chapter 10.

KILROY FCC — kernel parallel

When CONFIG_RTX_FIELD_DIE kernel stack runs, scale 0–1,000,000 μ from overshoot; entropy feedback clamps aggressive modes. Userspace guard and kernel FCC share vocabulary — Chapter 21 Queen + KILROY sovereign field. Ring transition does not reset operator literacy — same constants, stronger enforcement boundary.

Failure modes — NaN theology

Disabled guard in a fork yields white noise HDR, NaN entropy, frozen spiderweb, lying FCC mirrors.
Recovery: restore harmonics guard, reset analog defaults, clear fabrics, new sealed session. Do not ship screenshots of accidents as [Visual](#) without label.

Figure 9.5 — NaN theology: FCC and CFL refuse dishonest steps.

Claim: Default ./linux.sh run boots Field Die — x86.comp path

Operator drills

Drill 9.A — CFL clamp

```
set AnalogFields.WaveSpeed 9.9
set AnalogFields.TimeScale 4.0
list AnalogFields
grep -iE 'cfl|clamp|scale' run.log
```

Drill 9.B — Tesla grep

```
grep -E 'TESLA_R_|FIELD_PHI_MILLI|data_bus'
Navigator/engine/FieldRtxFieldAbs.hpp
```

Drill 9.C — Propalactic honesty

```
set AnalogFields.PropalacticScale 2.0
# Observe Phi forcing – label dynamics knob in operator notes
```

Chapter summary

CFL guard enforces wave and diffusion limits before GPU evolution. FCC floats pack to data_bus[16–23]. Tesla constants bias direction — slots 31, 34. KILROY extends discipline to kernel. PropalacticScale and GateFidelity are powerful — grep and label.

Prior: Chapter 8 Next: Chapter 10

Study questions

1. Write CFL inequalities and explain Δx^2 .
2. What happens when $\text{waveCFL} > 1$ before dispatch?
3. Quote Tesla forward/reverse resistance.
4. Which slots publish Tesla bias?
5. How is PropalacticScale labeled in Chapter 12?
6. What is KILROY FCC overshoot scale?
7. Why does resolution upscale interact with thermoCFL?
8. Name three FCC floats and stability roles.
9. What is NaN theology?
10. How does Tesla affect spiderweb edges?

Coupling energy — Maxwell neighborhood on FCC bus

FieldCoupling slot 23 is the dial Chapter 15 credits to Maxwell's insight: neighbors exchange state. Raising coupling without CFL violation still raises field work — ThermoAccountant entropyThisFrame responds. Operators tuning coupling should grep THERMO and slot 23 together across ten dispatches.

InjectStrength and probe ethics

Mouse probes inject energy — InjectStrength slot 21. CFL guard may clamp inject when coupled with high TimeScale. Offense with probes is operator consent to move energy; document sessions when teaching.

Cross-chapter capstone — FCC to fabric to spiderweb

FCC admitted post-guard → bindings 8-10 evolve → updateHardwareFromAnalogFields mirrors → list Hardware. One chain, three witnesses. Chapter 12 says label each.

Case study 1 — operator adjusts FCC under load

Pedagogical scenario 1 for harmonics internalization.

Scenario setup

Operator session 1 begins on default x86 die. Sealed time T0. Resolution tier moderate. Baseline WaveSpeed 1.0, TimeScale 1.0, ThermoAlpha default, FieldCoupling 0.5.

Operator increases TimeScale by 0.3 steps until STATUS shows entropy shift or grep hints clamp. Record admitted WaveSpeed and slot 18 mirror at each step.

Interpretation

If THERMO rises without CFL message, coupling or inject may dominate — not every entropy change is CFL.

If list AnalogFields shows lower WaveSpeed than set command requested, harmonics guard admitted lower value — document requested vs admitted in operator journal.

Tesla witness

After FCC stable, toggle TeslaBiasStrength if exposed in build. Watch data_bus 31/34 and spiderweb edge util in list Hardware — directional story should appear without claiming fluidics lab measurement.

Case study 2 — operator adjusts FCC under load

Pedagogical scenario 2 for harmonics internalization.

Scenario setup

Operator session 2 begins on default x86 die. Sealed time T1. Resolution tier moderate. Baseline WaveSpeed 1.0, TimeScale 1.0, ThermoAlpha default, FieldCoupling 0.5.

Operator increases TimeScale by 0.3 steps until STATUS shows entropy shift or grep hints clamp. Record admitted WaveSpeed and slot 18 mirror at each step.

Interpretation

If THERMO rises without CFL message, coupling or inject may dominate — not every entropy change is CFL.

If list AnalogFields shows lower WaveSpeed than set command requested, harmonics guard admitted lower value — document requested vs admitted in operator journal.

Tesla witness

After FCC stable, toggle TeslaBiasStrength if exposed in build. Watch data_bus 31/34 and spiderweb edge util in list Hardware — directional story should appear without claiming fluidics lab measurement.

Case study 3 — operator adjusts FCC under load

Pedagogical scenario 3 for harmonics internalization.

Scenario setup

Operator session 3 begins on default x86 die. Sealed time T2. Resolution tier moderate. Baseline WaveSpeed 1.0, TimeScale 1.0, ThermoAlpha default, FieldCoupling 0.5.

Operator increases TimeScale by 0.3 steps until STATUS shows entropy shift or grep hints clamp. Record admitted WaveSpeed and slot 18 mirror at each step.

Interpretation

If THERMO rises without CFL message, coupling or inject may dominate — not every entropy change is CFL.

If list AnalogFields shows lower WaveSpeed than set command requested, harmonics guard admitted lower value — document requested vs admitted in operator journal.

Tesla witness

After FCC stable, toggle TeslaBiasStrength if exposed in build. Watch data_bus 31/34 and spiderweb edge util in list Hardware — directional story should appear without claiming fluidics lab measurement.

Case study 4 — operator adjusts FCC under load

Pedagogical scenario 4 for harmonics internalization.

Scenario setup

Operator session 4 begins on default x86 die. Sealed time T3. Resolution tier moderate. Baseline WaveSpeed 1.0, TimeScale 1.0, ThermoAlpha default, FieldCoupling 0.5.

Operator increases TimeScale by 0.3 steps until STATUS shows entropy shift or grep hints clamp. Record admitted WaveSpeed and slot 18 mirror at each step.

Interpretation

If THERMO rises without CFL message, coupling or inject may dominate — not every entropy change is CFL.

If list AnalogFields shows lower WaveSpeed than set command requested, harmonics guard admitted lower value — document requested vs admitted in operator journal.

Tesla witness

After FCC stable, toggle TeslaBiasStrength if exposed in build. Watch data_bus 31/34 and spiderweb edge util in list Hardware — directional story should appear without claiming fluidics lab measurement.

Case study 5 — operator adjusts FCC under load

Pedagogical scenario 5 for harmonics internalization.

Scenario setup

Operator session 5 begins on default x86 die. Sealed time T4. Resolution tier moderate. Baseline WaveSpeed 1.0, TimeScale 1.0, ThermoAlpha default, FieldCoupling 0.5.

Operator increases TimeScale by 0.3 steps until STATUS shows entropy shift or grep hints clamp. Record admitted WaveSpeed and slot 18 mirror at each step.

Interpretation

If THERMO rises without CFL message, coupling or inject may dominate — not every entropy change is CFL.

If list AnalogFields shows lower WaveSpeed than set command requested, harmonics guard admitted lower value — document requested vs admitted in operator journal.

Tesla witness

After FCC stable, toggle TeslaBiasStrength if exposed in build. Watch data_bus 31/34 and spiderweb edge util in list Hardware — directional story should appear without claiming fluidics lab measurement.

Case study 6 — operator adjusts FCC under load

Pedagogical scenario 6 for harmonics internalization.

Scenario setup

Operator session 6 begins on default x86 die. Sealed time T5. Resolution tier moderate. Baseline WaveSpeed 1.0, TimeScale 1.0, ThermoAlpha default, FieldCoupling 0.5.

Operator increases TimeScale by 0.3 steps until STATUS shows entropy shift or grep hints clamp. Record admitted WaveSpeed and slot 18 mirror at each step.

Interpretation

If THERMO rises without CFL message, coupling or inject may dominate — not every entropy change is CFL.

If list AnalogFields shows lower WaveSpeed than set command requested, harmonics guard admitted lower value — document requested vs admitted in operator journal.

Tesla witness

After FCC stable, toggle TeslaBiasStrength if exposed in build. Watch data_bus 31/34 and spiderweb edge util in list Hardware — directional story should appear without claiming fluidics lab measurement.

Case study 7 — operator adjusts FCC under load

Pedagogical scenario 7 for harmonics internalization.

Scenario setup

Operator session 7 begins on default x86 die. Sealed time T6. Resolution tier moderate. Baseline WaveSpeed 1.0, TimeScale 1.0, ThermoAlpha default, FieldCoupling 0.5.

Operator increases TimeScale by 0.3 steps until STATUS shows entropy shift or grep hints clamp. Record admitted WaveSpeed and slot 18 mirror at each step.

Interpretation

If THERMO rises without CFL message, coupling or inject may dominate — not every entropy change is CFL.

If list AnalogFields shows lower WaveSpeed than set command requested, harmonics guard admitted lower value — document requested vs admitted in operator journal.

Tesla witness

After FCC stable, toggle TeslaBiasStrength if exposed in build. Watch data_bus 31/34 and spiderweb edge util in list Hardware — directional story should appear without claiming fluidics lab measurement.

Case study 8 — operator adjusts FCC under load

Pedagogical scenario 8 for harmonics internalization.

Scenario setup

Operator session 8 begins on default x86 die. Sealed time T7. Resolution tier moderate. Baseline WaveSpeed 1.0, TimeScale 1.0, ThermoAlpha default, FieldCoupling 0.5.

Operator increases TimeScale by 0.3 steps until STATUS shows entropy shift or grep hints clamp. Record admitted WaveSpeed and slot 18 mirror at each step.

Interpretation

If THERMO rises without CFL message, coupling or inject may dominate — not every entropy change is CFL.

If list AnalogFields shows lower WaveSpeed than set command requested, harmonics guard admitted lower value — document requested vs admitted in operator journal.

Tesla witness

After FCC stable, toggle TeslaBiasStrength if exposed in build. Watch data_bus 31/34 and spiderweb edge util in list Hardware — directional story should appear without claiming fluidics lab measurement.

Case study 9 — operator adjusts FCC under load

Pedagogical scenario 9 for harmonics internalization.

Scenario setup

Operator session 9 begins on default x86 die. Sealed time T8. Resolution tier moderate. Baseline WaveSpeed 1.0, TimeScale 1.0, ThermoAlpha default, FieldCoupling 0.5.

Operator increases TimeScale by 0.3 steps until STATUS shows entropy shift or grep hints clamp. Record admitted WaveSpeed and slot 18 mirror at each step.

Interpretation

If THERMO rises without CFL message, coupling or inject may dominate — not every entropy change is CFL.

If list AnalogFields shows lower WaveSpeed than set command requested, harmonics guard admitted lower value — document requested vs admitted in operator journal.

Tesla witness

After FCC stable, toggle TeslaBiasStrength if exposed in build. Watch data_bus 31/34 and spiderweb edge util in list Hardware — directional story should appear without claiming fluidics lab measurement.

Case study 10 — operator adjusts FCC under load

Pedagogical scenario 10 for harmonics internalization.

Scenario setup

Operator session 10 begins on default x86 die. Sealed time T9. Resolution tier moderate. Baseline WaveSpeed 1.0, TimeScale 1.0, ThermoAlpha default, FieldCoupling 0.5.

Operator increases TimeScale by 0.3 steps until STATUS shows entropy shift or grep hints clamp. Record admitted WaveSpeed and slot 18 mirror at each step.

Interpretation

If THERMO rises without CFL message, coupling or inject may dominate — not every entropy change is CFL.

If list AnalogFields shows lower WaveSpeed than set command requested, harmonics guard admitted lower value — document requested vs admitted in operator journal.

Tesla witness

After FCC stable, toggle TeslaBiasStrength if exposed in build. Watch data_bus 31/34 and spiderweb edge util in list Hardware — directional story should appear without claiming fluidics lab measurement.

Case study 11 — operator adjusts FCC under load

Pedagogical scenario 11 for harmonics internalization.

Scenario setup

Operator session 11 begins on default x86 die. Sealed time T10. Resolution tier moderate. Baseline WaveSpeed 1.0, TimeScale 1.0, ThermoAlpha default, FieldCoupling 0.5.

Operator increases TimeScale by 0.3 steps until STATUS shows entropy shift or grep hints clamp. Record admitted WaveSpeed and slot 18 mirror at each step.

Interpretation

If THERMO rises without CFL message, coupling or inject may dominate — not every entropy change is CFL.

If list AnalogFields shows lower WaveSpeed than set command requested, harmonics guard admitted lower value — document requested vs admitted in operator journal.

Tesla witness

After FCC stable, toggle TeslaBiasStrength if exposed in build. Watch data_bus 31/34 and spiderweb edge util in list Hardware — directional story should appear without claiming fluidics lab measurement.

Case study 12 — operator adjusts FCC under load

Pedagogical scenario 12 for harmonics internalization.

Scenario setup

Operator session 12 begins on default x86 die. Sealed time T11. Resolution tier moderate. Baseline WaveSpeed 1.0, TimeScale 1.0, ThermoAlpha default, FieldCoupling 0.5.

Operator increases TimeScale by 0.3 steps until STATUS shows entropy shift or grep hints clamp. Record admitted WaveSpeed and slot 18 mirror at each step.

Interpretation

If THERMO rises without CFL message, coupling or inject may dominate — not every entropy change is CFL.

If list AnalogFields shows lower WaveSpeed than set command requested, harmonics guard admitted lower value — document requested vs admitted in operator journal.

Tesla witness

After FCC stable, toggle TeslaBiasStrength if exposed in build. Watch data_bus 31/34 and spiderweb edge util in list Hardware — directional story should appear without claiming fluidics lab measurement.

Case study 13 — operator adjusts FCC under load

Pedagogical scenario 13 for harmonics internalization.

Scenario setup

Operator session 13 begins on default x86 die. Sealed time T12. Resolution tier moderate. Baseline WaveSpeed 1.0, TimeScale 1.0, ThermoAlpha default, FieldCoupling 0.5.

Operator increases TimeScale by 0.3 steps until STATUS shows entropy shift or grep hints clamp. Record admitted WaveSpeed and slot 18 mirror at each step.

Interpretation

If THERMO rises without CFL message, coupling or inject may dominate — not every entropy change is CFL.

If list AnalogFields shows lower WaveSpeed than set command requested, harmonics guard admitted lower value — document requested vs admitted in operator journal.

Tesla witness

After FCC stable, toggle TeslaBiasStrength if exposed in build. Watch data_bus 31/34 and spiderweb edge util in list Hardware — directional story should appear without claiming fluidics lab measurement.

Case study 14 — operator adjusts FCC under load

Pedagogical scenario 14 for harmonics internalization.

Scenario setup

Operator session 14 begins on default x86 die. Sealed time T13. Resolution tier moderate. Baseline WaveSpeed 1.0, TimeScale 1.0, ThermoAlpha default, FieldCoupling 0.5.

Operator increases TimeScale by 0.3 steps until STATUS shows entropy shift or grep hints clamp. Record admitted WaveSpeed and slot 18 mirror at each step.

Interpretation

If THERMO rises without CFL message, coupling or inject may dominate — not every entropy change is CFL.

If list AnalogFields shows lower WaveSpeed than set command requested, harmonics guard admitted lower value — document requested vs admitted in operator journal.

Tesla witness

After FCC stable, toggle TeslaBiasStrength if exposed in build. Watch data_bus 31/34 and spiderweb edge util in list Hardware — directional story should appear without claiming fluidics lab measurement.

Case study 15 — operator adjusts FCC under load

Pedagogical scenario 15 for harmonics internalization.

Scenario setup

Operator session 15 begins on default x86 die. Sealed time T14. Resolution tier moderate. Baseline WaveSpeed 1.0, TimeScale 1.0, ThermoAlpha default, FieldCoupling 0.5.

Operator increases TimeScale by 0.3 steps until STATUS shows entropy shift or grep hints clamp. Record admitted WaveSpeed and slot 18 mirror at each step.

Interpretation

If THERMO rises without CFL message, coupling or inject may dominate — not every entropy change is CFL.

If list AnalogFields shows lower WaveSpeed than set command requested, harmonics guard admitted lower value — document requested vs admitted in operator journal.

Tesla witness

After FCC stable, toggle TeslaBiasStrength if exposed in build. Watch data_bus 31/34 and spiderweb edge util in list Hardware — directional story should appear without claiming fluidics lab measurement.

Case study 16 — operator adjusts FCC under load

Pedagogical scenario 16 for harmonics internalization.

Scenario setup

Operator session 16 begins on default x86 die. Sealed time T15. Resolution tier moderate. Baseline WaveSpeed 1.0, TimeScale 1.0, ThermoAlpha default, FieldCoupling 0.5.

Operator increases TimeScale by 0.3 steps until STATUS shows entropy shift or grep hints clamp. Record admitted WaveSpeed and slot 18 mirror at each step.

Interpretation

If THERMO rises without CFL message, coupling or inject may dominate — not every entropy change is CFL.

If list AnalogFields shows lower WaveSpeed than set command requested, harmonics guard admitted lower value — document requested vs admitted in operator journal.

Tesla witness

After FCC stable, toggle TeslaBiasStrength if exposed in build. Watch data_bus 31/34 and spiderweb edge util in list Hardware — directional story should appear without claiming fluidics lab measurement.

Case study 17 — operator adjusts FCC under load

Pedagogical scenario 17 for harmonics internalization.

Scenario setup

Operator session 17 begins on default x86 die. Sealed time T16. Resolution tier moderate. Baseline WaveSpeed 1.0, TimeScale 1.0, ThermoAlpha default, FieldCoupling 0.5.

Operator increases TimeScale by 0.3 steps until STATUS shows entropy shift or grep hints clamp. Record admitted WaveSpeed and slot 18 mirror at each step.

Interpretation

If THERMO rises without CFL message, coupling or inject may dominate — not every entropy change is CFL.

If list AnalogFields shows lower WaveSpeed than set command requested, harmonics guard admitted lower value — document requested vs admitted in operator journal.

Tesla witness

After FCC stable, toggle TeslaBiasStrength if exposed in build. Watch data_bus 31/34 and spiderweb edge util in list Hardware — directional story should appear without claiming fluidics lab measurement.

Case study 18 — operator adjusts FCC under load

Pedagogical scenario 18 for harmonics internalization.

Scenario setup

Operator session 18 begins on default x86 die. Sealed time T17. Resolution tier moderate. Baseline WaveSpeed 1.0, TimeScale 1.0, ThermoAlpha default, FieldCoupling 0.5.

Operator increases TimeScale by 0.3 steps until STATUS shows entropy shift or grep hints clamp. Record admitted WaveSpeed and slot 18 mirror at each step.

Interpretation

If THERMO rises without CFL message, coupling or inject may dominate — not every entropy change is CFL.

If list AnalogFields shows lower WaveSpeed than set command requested, harmonics guard admitted lower value — document requested vs admitted in operator journal.

Tesla witness

After FCC stable, toggle TeslaBiasStrength if exposed in build. Watch data_bus 31/34 and spiderweb edge util in list Hardware — directional story should appear without claiming fluidics lab measurement.

Case study 19 — operator adjusts FCC under load

Pedagogical scenario 19 for harmonics internalization.

Scenario setup

Operator session 19 begins on default x86 die. Sealed time T18. Resolution tier moderate. Baseline WaveSpeed 1.0, TimeScale 1.0, ThermoAlpha default, FieldCoupling 0.5.

Operator increases TimeScale by 0.3 steps until STATUS shows entropy shift or grep hints clamp. Record admitted WaveSpeed and slot 18 mirror at each step.

Interpretation

If THERMO rises without CFL message, coupling or inject may dominate — not every entropy change is CFL.

If list AnalogFields shows lower WaveSpeed than set command requested, harmonics guard admitted lower value — document requested vs admitted in operator journal.

Tesla witness

After FCC stable, toggle TeslaBiasStrength if exposed in build. Watch data_bus 31/34 and spiderweb edge util in list Hardware — directional story should appear without claiming fluidics lab measurement.

Case study 20 — operator adjusts FCC under load

Pedagogical scenario 20 for harmonics internalization.

Scenario setup

Operator session 20 begins on default x86 die. Sealed time T19. Resolution tier moderate. Baseline WaveSpeed 1.0, TimeScale 1.0, ThermoAlpha default, FieldCoupling 0.5.

Operator increases TimeScale by 0.3 steps until STATUS shows entropy shift or grep hints clamp. Record admitted WaveSpeed and slot 18 mirror at each step.

Interpretation

If THERMO rises without CFL message, coupling or inject may dominate — not every entropy change is CFL.

If list AnalogFields shows lower WaveSpeed than set command requested, harmonics guard admitted lower value — document requested vs admitted in operator journal.

Tesla witness

After FCC stable, toggle TeslaBiasStrength if exposed in build. Watch data_bus 31/34 and spiderweb edge util in list Hardware — directional story should appear without claiming fluidics lab measurement.

Case study 21 — operator adjusts FCC under load

Pedagogical scenario 21 for harmonics internalization.

Scenario setup

Operator session 21 begins on default x86 die. Sealed time T20. Resolution tier moderate. Baseline WaveSpeed 1.0, TimeScale 1.0, ThermoAlpha default, FieldCoupling 0.5.

Operator increases TimeScale by 0.3 steps until STATUS shows entropy shift or grep hints clamp. Record admitted WaveSpeed and slot 18 mirror at each step.

Interpretation

If THERMO rises without CFL message, coupling or inject may dominate — not every entropy change is CFL.

If list AnalogFields shows lower WaveSpeed than set command requested, harmonics guard admitted lower value — document requested vs admitted in operator journal.

Tesla witness

After FCC stable, toggle TeslaBiasStrength if exposed in build. Watch data_bus 31/34 and spiderweb edge util in list Hardware — directional story should appear without claiming fluidics lab measurement.

Case study 22 — operator adjusts FCC under load

Pedagogical scenario 22 for harmonics internalization.

Scenario setup

Operator session 22 begins on default x86 die. Sealed time T21. Resolution tier moderate. Baseline WaveSpeed 1.0, TimeScale 1.0, ThermoAlpha default, FieldCoupling 0.5.

Operator increases TimeScale by 0.3 steps until STATUS shows entropy shift or grep hints clamp. Record admitted WaveSpeed and slot 18 mirror at each step.

Interpretation

If THERMO rises without CFL message, coupling or inject may dominate — not every entropy change is CFL.

If list AnalogFields shows lower WaveSpeed than set command requested, harmonics guard admitted lower value — document requested vs admitted in operator journal.

Tesla witness

After FCC stable, toggle TeslaBiasStrength if exposed in build. Watch data_bus 31/34 and spiderweb edge util in list Hardware — directional story should appear without claiming fluidics lab measurement.

Case study 23 — operator adjusts FCC under load

Pedagogical scenario 23 for harmonics internalization.

Scenario setup

Operator session 23 begins on default x86 die. Sealed time T22. Resolution tier moderate. Baseline WaveSpeed 1.0, TimeScale 1.0, ThermoAlpha default, FieldCoupling 0.5.

Operator increases TimeScale by 0.3 steps until STATUS shows entropy shift or grep hints clamp. Record admitted WaveSpeed and slot 18 mirror at each step.

Interpretation

If THERMO rises without CFL message, coupling or inject may dominate — not every entropy change is CFL.

If list AnalogFields shows lower WaveSpeed than set command requested, harmonics guard admitted lower value — document requested vs admitted in operator journal.

Tesla witness

After FCC stable, toggle TeslaBiasStrength if exposed in build. Watch data_bus 31/34 and spiderweb edge util in list Hardware — directional story should appear without claiming fluidics lab measurement.

Case study 24 — operator adjusts FCC under load

Pedagogical scenario 24 for harmonics internalization.

Scenario setup

Operator session 24 begins on default x86 die. Sealed time T23. Resolution tier moderate. Baseline WaveSpeed 1.0, TimeScale 1.0, ThermoAlpha default, FieldCoupling 0.5.

Operator increases TimeScale by 0.3 steps until STATUS shows entropy shift or grep hints clamp. Record admitted WaveSpeed and slot 18 mirror at each step.

Interpretation

If THERMO rises without CFL message, coupling or inject may dominate — not every entropy change is CFL.

If list AnalogFields shows lower WaveSpeed than set command requested, harmonics guard admitted lower value — document requested vs admitted in operator journal.

Tesla witness

After FCC stable, toggle TeslaBiasStrength if exposed in build. Watch data_bus 31/34 and spiderweb edge util in list Hardware — directional story should appear without claiming fluidics lab measurement.

Case study 25 — operator adjusts FCC under load

Pedagogical scenario 25 for harmonics internalization.

Scenario setup

Operator session 25 begins on default x86 die. Sealed time T24. Resolution tier moderate. Baseline WaveSpeed 1.0, TimeScale 1.0, ThermoAlpha default, FieldCoupling 0.5.

Operator increases TimeScale by 0.3 steps until STATUS shows entropy shift or grep hints clamp. Record admitted WaveSpeed and slot 18 mirror at each step.

Interpretation

If THERMO rises without CFL message, coupling or inject may dominate — not every entropy change is CFL.

If list AnalogFields shows lower WaveSpeed than set command requested, harmonics guard admitted lower value — document requested vs admitted in operator journal.

Tesla witness

After FCC stable, toggle TeslaBiasStrength if exposed in build. Watch data_bus 31/34 and spiderweb edge util in list Hardware — directional story should appear without claiming fluidics lab measurement.

Case study 26 — operator adjusts FCC under load

Pedagogical scenario 26 for harmonics internalization.

Scenario setup

Operator session 26 begins on default x86 die. Sealed time T25. Resolution tier moderate. Baseline WaveSpeed 1.0, TimeScale 1.0, ThermoAlpha default, FieldCoupling 0.5.

Operator increases TimeScale by 0.3 steps until STATUS shows entropy shift or grep hints clamp. Record admitted WaveSpeed and slot 18 mirror at each step.

Interpretation

If THERMO rises without CFL message, coupling or inject may dominate — not every entropy change is CFL.

If list AnalogFields shows lower WaveSpeed than set command requested, harmonics guard admitted lower value — document requested vs admitted in operator journal.

Tesla witness

After FCC stable, toggle TeslaBiasStrength if exposed in build. Watch data_bus 31/34 and spiderweb edge util in list Hardware — directional story should appear without claiming fluidics lab measurement.

Case study 27 — operator adjusts FCC under load

Pedagogical scenario 27 for harmonics internalization.

Scenario setup

Operator session 27 begins on default x86 die. Sealed time T26. Resolution tier moderate. Baseline WaveSpeed 1.0, TimeScale 1.0, ThermoAlpha default, FieldCoupling 0.5.

Operator increases TimeScale by 0.3 steps until STATUS shows entropy shift or grep hints clamp. Record admitted WaveSpeed and slot 18 mirror at each step.

Interpretation

If THERMO rises without CFL message, coupling or inject may dominate — not every entropy change is CFL.

If list AnalogFields shows lower WaveSpeed than set command requested, harmonics guard admitted lower value — document requested vs admitted in operator journal.

Tesla witness

After FCC stable, toggle TeslaBiasStrength if exposed in build. Watch data_bus 31/34 and spiderweb edge util in list Hardware — directional story should appear without claiming fluidics lab measurement.

Case study 28 — operator adjusts FCC under load

Pedagogical scenario 28 for harmonics internalization.

Scenario setup

Operator session 28 begins on default x86 die. Sealed time T27. Resolution tier moderate. Baseline WaveSpeed 1.0, TimeScale 1.0, ThermoAlpha default, FieldCoupling 0.5.

Operator increases TimeScale by 0.3 steps until STATUS shows entropy shift or grep hints clamp. Record admitted WaveSpeed and slot 18 mirror at each step.

Interpretation

If THERMO rises without CFL message, coupling or inject may dominate — not every entropy change is CFL.

If list AnalogFields shows lower WaveSpeed than set command requested, harmonics guard admitted lower value — document requested vs admitted in operator journal.

Tesla witness

After FCC stable, toggle TeslaBiasStrength if exposed in build. Watch data_bus 31/34 and spiderweb edge util in list Hardware — directional story should appear without claiming fluidics lab measurement.

Evidence anchor — grep and sources

Major claims in this chapter anchored for reproducibility. Implemented = grep today; Metaphor
= intuition; Philosophy = discipline.

Claim	Statement	Label	Evidence
Wave CFL	Stability before step	Implemented	Host guard scales unsafe dt
Tesla bias	Directional preference	Metaphor	TESLA_R_* → data_bus 31/34
FCC bus slots	Telemetry floats	Implemented	data_bus[16–23]
Fluidic valve	Physical Tesla part	Metaphor	Not in chassis

```
forward ease / reverse drag via TESLA_R_* constants
```

```
grep -E 'TESLA_R_|waveCFL' Navigator/engine/
```

Source paths

- Navigator/engine/HarmonicsGuard.cpp
- Navigator/engine/FCC.cpp

Chapter summary — before you turn the page

CFL and Tesla bias are numerical ethics before fabric evolution — forward ease, reverse drag, host refuses explosion. KILROY FCC parallels userspace guards.

Hardware Mirror — Spiderweb

CHAPTER 10 · HARDWARE SPIDERWEB — SUB-MICRON MIRROR

Learning objectives

- Explain `hardwareFabric` as read-only mirror of fabric averages.
- Trace `updateHardwareFromAnalogFields()` six-step frame ritual.
- Define `voltageFactor`, `thermalThrottle`, `parallelEff` roles.
- Operate mastery tiers `Puny`, `Adept`, `Tidewalker` with honest expectations.
- Apply sub-micron honesty table before marketing language.

IN THIS CHAPTER

- | | |
|---|--|
| 1. Introduction — dashboard, not microscope | 8. Chapter summary |
| 2. Six-step frame ritual | 9. Study questions |
| 3. Derived factors | 10. Coupling fabric averages to sysfs <code>Puny</code> tier |
| 4. Mastery tiers | 11. <code>SimulateSubMicron</code> — what changes |
| 5. Sub-micron honesty table | 12. <code>Tidewalker</code> responsibility |
| 6. <code>precision-field.py</code> — NEXUS cousin | 13. Mirror anatomy 1 — units, edges, and sysfs |
| 7. Operator drills | 14. Mirror anatomy 2 — units, edges, and sysfs |

On the way — what you will learn

The spiderweb is a **read-only mirror** — fabric averages compressed for the operator dashboard, not a second simulation. On the way you will follow `updateHardwareFromAnalogFields()`, read per-core MHz and util, and refuse SEM marketing as instrumentation.

On the way — Spiderweb graph

- voltageFactor, thermalThrottle, parallelEff
- Puny tier sysfs witness
- Lossy pipeline: averages hide texels
- list Hardware prompt ritual

Introduction — dashboard, not microscope

`updateHardwareFromAnalogFields()` mirrors averaged Phi/Thermo/Flow into `hardwareFabric` — per-core MHz, util, temp, power, spiderweb edge currents. Read-only mirror — not second simulation, not SEM imaging. Chapter 9 Tesla constants damp reverse edges; Chapter 7 dispatch calls mirror every frame after fabric evolution.

Figure 10.1 — Fabric averages drive util graph.

Six-step frame ritual

Figure 10.2 — Fabric averages compress into spiderweb util — lossy mirror.

1. Sample avg Phi, Thermo, Flow
2. Apply fluid velocity/density + Tesla bias
3. Compute voltageFactor, thermalThrottle, parallelEff
4. Update `hardwareFabric.units[]` — `operationalFreqMHz`, util, voltage, temp, power
5. Update spiderweb `edges[].currentUtil`
6. Accumulate `simulatedChipCycles`

Derived factors

Factor	Story	Label
voltageFactor	Phi-linked electrical metaphor	Metaphor + fabric witness
thermalThrottle	Thermo-linked heat story	Proxy
parallelEff	Flow-linked utilization shape	Proxy

Mastery tiers

Tier	Controls
Puny	ShowInStatusLog, AutoUseRealClocks — sysfs MHz Linux
Adept	TargetCoreClockMHz, ThermalSensitivity, SimulateSubMicron
Tidewalker	EnableSpiderwebGraph, ForcedVendor, SubMicronDetail

Sub-micron honesty table

Claim	Reality	Label
Adaptive 320×200 → 4K+	Implemented	Implemented
SDF epsilons + accumulation	Implemented	Implemented
Zero-cost SEM fidelity	Procedural pixel detail	Metaphor

precision-field.py — NEXUS cousin

GPS-anchored entity map, spiderweb nodes, thermal-earth bodies — separate codebase, cousin metaphor.
Correlate at operator mind with engine spiderweb, not automatic merge.

Figure 10.3 — Dispatch thermo feeds spiderweb averages — know the compression.

Claim: Thin host → vkCmdDispatch — offensive write each tick

Figure 10.4 — data_bus witness beside spiderweb mirror — honesty table.

Claim: data_bus[64] telemetry spine per dispatch

Operator drills

Drill 10.A

```
list Hardware
set Hardware.SimulateSubMicron 1
grep -i spiderweb run.log
```

Chapter summary

Spiderweb mirrors fabric. Six steps each frame. Tiers unlock controls. SEM is metaphor. Prior: Ch 9 Next: Ch 11.

Study questions

1. List six mirror steps.
2. What does Puny tier read for clocks?
3. Label SimulateSubMicron honesty.
4. How does Tesla reach edges?
5. What is precision-field cousin?

Coupling fabric averages to sysfs Puny tier

Puny AutoUseRealClocks reads Linux sysfs MHz as freq witness — Chapter 19 sovereign time extends witness doctrine. Spiderweb simulated MHz should not be confused with sysfs — honesty table: mirror vs hardware read.

SimulateSubMicron — what changes

Adept tier SimulateSubMicron enables procedural detail path — still not SEM. SDF epsilons accumulate; label Metaphor when presenting screenshots externally.

Tidewalker responsibility

Graph override is power — Chapter 18 covenant: operator conscience, not daemon. ForcedVendor and SubMicronDetail can lie beautifully; stderr and sysfs Puny tier anchor honesty.

Mirror anatomy 1 — units, edges, and sysfs

Depth pass 1 on hardwareFabric literacy.

Per-core units[]

Each unit entry tracks operationalFreqMHz, utilization subfunctions, voltage, temperature, power proxy. Entry 0 is not a physical core map — it is dashboard metaphor aligned to sysfs when Puny AutoUseRealClocks reads real MHz.

Compare list Hardware output to grep STATUS — numbers should move together when fabric injected.

Spiderweb edges

Edges carry currentUtil as fabric-driven story. Tesla reverse damping from Chapter 9 modifies edge update — forward ease, reverse resist.

Tidewalker tier EnableSpiderwebGraph allows override — operator responsibility rises; Chapter 18 covenant applies.

simulatedChipCycles

Accumulated cycles are comparative telemetry — useful for session shape, not billing. Headless CI can track monotonic increase as dispatch health.

Mirror anatomy 2 — units, edges, and sysfs

Depth pass 2 on hardwareFabric literacy.

Per-core units[]

Each unit entry tracks operationalFreqMHz, utilization subfunctions, voltage, temperature, power proxy. Entry 1 is not a physical core map — it is dashboard metaphor aligned to sysfs when Puny AutoUseRealClocks reads real MHz.

Compare list Hardware output to grep STATUS — numbers should move together when fabric injected.

Spiderweb edges

Edges carry currentUtil as fabric-driven story. Tesla reverse damping from Chapter 9 modifies edge update — forward ease, reverse resist.

Tidewalker tier EnableSpiderwebGraph allows override — operator responsibility rises; Chapter 18 covenant applies.

simulatedChipCycles

Accumulated cycles are comparative telemetry — useful for session shape, not billing. Headless CI can track monotonic increase as dispatch health.

Mirror anatomy 3 — units, edges, and sysfs

Depth pass 3 on hardwareFabric literacy.

Per-core units[]

Each unit entry tracks operationalFreqMHz, utilization subfunctions, voltage, temperature, power proxy. Entry 2 is not a physical core map — it is dashboard metaphor aligned to sysfs when Puny AutoUseRealClocks reads real MHz.

Compare list Hardware output to grep STATUS — numbers should move together when fabric injected.

Spiderweb edges

Edges carry currentUtil as fabric-driven story. Tesla reverse damping from Chapter 9 modifies edge update — forward ease, reverse resist.

Tidewalker tier EnableSpiderwebGraph allows override — operator responsibility rises; Chapter 18 covenant applies.

simulatedChipCycles

Accumulated cycles are comparative telemetry — useful for session shape, not billing. Headless CI can track monotonic increase as dispatch health.

Mirror anatomy 4 — units, edges, and sysfs

Depth pass 4 on hardwareFabric literacy.

Per-core units[]

Each unit entry tracks operationalFreqMHz, utilization subfunctions, voltage, temperature, power proxy. Entry 3 is not a physical core map — it is dashboard metaphor aligned to sysfs when Puny AutoUseRealClocks reads real MHz.

Compare list Hardware output to grep STATUS — numbers should move together when fabric injected.

Spiderweb edges

Edges carry currentUtil as fabric-driven story. Tesla reverse damping from Chapter 9 modifies edge update — forward ease, reverse resist.

Tidewalker tier EnableSpiderwebGraph allows override — operator responsibility rises; Chapter 18 covenant applies.

simulatedChipCycles

Accumulated cycles are comparative telemetry — useful for session shape, not billing. Headless CI can track monotonic increase as dispatch health.

Mirror anatomy 5 — units, edges, and sysfs

Depth pass 5 on hardwareFabric literacy.

Per-core units[]

Each unit entry tracks operationalFreqMHz, utilization subfunctions, voltage, temperature, power proxy. Entry 4 is not a physical core map — it is dashboard metaphor aligned to sysfs when Puny AutoUseRealClocks reads real MHz.

Compare list Hardware output to grep STATUS — numbers should move together when fabric injected.

Spiderweb edges

Edges carry currentUtil as fabric-driven story. Tesla reverse damping from Chapter 9 modifies edge update — forward ease, reverse resist.

Tidewalker tier EnableSpiderwebGraph allows override — operator responsibility rises; Chapter 18 covenant applies.

simulatedChipCycles

Accumulated cycles are comparative telemetry — useful for session shape, not billing. Headless CI can track monotonic increase as dispatch health.

Mirror anatomy 6 — units, edges, and sysfs

Depth pass 6 on hardwareFabric literacy.

Per-core units[]

Each unit entry tracks operationalFreqMHz, utilization subfunctions, voltage, temperature, power proxy.
Entry 5 is not a physical core map — it is dashboard metaphor aligned to sysfs when Puny
AutoUseRealClocks reads real MHz.

Compare list Hardware output to grep STATUS — numbers should move together when fabric injected.

Spiderweb edges

Edges carry currentUtil as fabric-driven story. Tesla reverse damping from Chapter 9 modifies edge update
— forward ease, reverse resist.

Tidewalker tier EnableSpiderwebGraph allows override — operator responsibility rises; Chapter 18
covenant applies.

simulatedChipCycles

Accumulated cycles are comparative telemetry — useful for session shape, not billing. Headless CI can
track monotonic increase as dispatch health.

Mirror anatomy 7 — units, edges, and sysfs

Depth pass 7 on hardwareFabric literacy.

Per-core units[]

Each unit entry tracks operationalFreqMHz, utilization subfunctions, voltage, temperature, power proxy.
Entry 6 is not a physical core map — it is dashboard metaphor aligned to sysfs when Puny
AutoUseRealClocks reads real MHz.

Compare list Hardware output to grep STATUS — numbers should move together when fabric injected.

Spiderweb edges

Edges carry currentUtil as fabric-driven story. Tesla reverse damping from Chapter 9 modifies edge update
— forward ease, reverse resist.

Tidewalker tier EnableSpiderwebGraph allows override — operator responsibility rises; Chapter 18
covenant applies.

simulatedChipCycles

Accumulated cycles are comparative telemetry — useful for session shape, not billing. Headless CI can
track monotonic increase as dispatch health.

Mirror anatomy 8 — units, edges, and sysfs

Depth pass 8 on hardwareFabric literacy.

Per-core units[]

Each unit entry tracks operationalFreqMHz, utilization subfunctions, voltage, temperature, power proxy. Entry 7 is not a physical core map — it is dashboard metaphor aligned to sysfs when Puny AutoUseRealClocks reads real MHz.

Compare list Hardware output to grep STATUS — numbers should move together when fabric injected.

Spiderweb edges

Edges carry currentUtil as fabric-driven story. Tesla reverse damping from Chapter 9 modifies edge update — forward ease, reverse resist.

Tidewalker tier EnableSpiderwebGraph allows override — operator responsibility rises; Chapter 18 covenant applies.

simulatedChipCycles

Accumulated cycles are comparative telemetry — useful for session shape, not billing. Headless CI can track monotonic increase as dispatch health.

Mirror anatomy 9 — units, edges, and sysfs

Depth pass 9 on hardwareFabric literacy.

Per-core units[]

Each unit entry tracks operationalFreqMHz, utilization subfunctions, voltage, temperature, power proxy. Entry 8 is not a physical core map — it is dashboard metaphor aligned to sysfs when Puny AutoUseRealClocks reads real MHz.

Compare list Hardware output to grep STATUS — numbers should move together when fabric injected.

Spiderweb edges

Edges carry currentUtil as fabric-driven story. Tesla reverse damping from Chapter 9 modifies edge update — forward ease, reverse resist.

Tidewalker tier EnableSpiderwebGraph allows override — operator responsibility rises; Chapter 18 covenant applies.

simulatedChipCycles

Accumulated cycles are comparative telemetry — useful for session shape, not billing. Headless CI can track monotonic increase as dispatch health.

Mirror anatomy 10 — units, edges, and sysfs

Depth pass 10 on hardwareFabric literacy.

Per-core units[]

Each unit entry tracks operationalFreqMHz, utilization subfunctions, voltage, temperature, power proxy. Entry 9 is not a physical core map — it is dashboard metaphor aligned to sysfs when Puny AutoUseRealClocks reads real MHz.

Compare list Hardware output to grep STATUS — numbers should move together when fabric injected.

Spiderweb edges

Edges carry currentUtil as fabric-driven story. Tesla reverse damping from Chapter 9 modifies edge update — forward ease, reverse resist.

Tidewalker tier EnableSpiderwebGraph allows override — operator responsibility rises; Chapter 18 covenant applies.

simulatedChipCycles

Accumulated cycles are comparative telemetry — useful for session shape, not billing. Headless CI can track monotonic increase as dispatch health.

Mirror anatomy 11 — units, edges, and sysfs

Depth pass 11 on hardwareFabric literacy.

Per-core units[]

Each unit entry tracks operationalFreqMHz, utilization subfunctions, voltage, temperature, power proxy. Entry 10 is not a physical core map — it is dashboard metaphor aligned to sysfs when Puny AutoUseRealClocks reads real MHz.

Compare list Hardware output to grep STATUS — numbers should move together when fabric injected.

Spiderweb edges

Edges carry currentUtil as fabric-driven story. Tesla reverse damping from Chapter 9 modifies edge update — forward ease, reverse resist.

Tidewalker tier EnableSpiderwebGraph allows override — operator responsibility rises; Chapter 18 covenant applies.

simulatedChipCycles

Accumulated cycles are comparative telemetry — useful for session shape, not billing. Headless CI can track monotonic increase as dispatch health.

Mirror anatomy 12 — units, edges, and sysfs

Depth pass 12 on hardwareFabric literacy.

Per-core units[]

Each unit entry tracks operationalFreqMHz, utilization subfunctions, voltage, temperature, power proxy. Entry 11 is not a physical core map — it is dashboard metaphor aligned to sysfs when Puny AutoUseRealClocks reads real MHz.

Compare list Hardware output to grep STATUS — numbers should move together when fabric injected.

Spiderweb edges

Edges carry currentUtil as fabric-driven story. Tesla reverse damping from Chapter 9 modifies edge update — forward ease, reverse resist.

Tidewalker tier EnableSpiderwebGraph allows override — operator responsibility rises; Chapter 18 covenant applies.

simulatedChipCycles

Accumulated cycles are comparative telemetry — useful for session shape, not billing. Headless CI can track monotonic increase as dispatch health.

Mirror anatomy 13 — units, edges, and sysfs

Depth pass 13 on hardwareFabric literacy.

Per-core units[]

Each unit entry tracks operationalFreqMHz, utilization subfunctions, voltage, temperature, power proxy. Entry 12 is not a physical core map — it is dashboard metaphor aligned to sysfs when Puny

AutoUseRealClocks reads real MHz.

Compare list Hardware output to grep STATUS — numbers should move together when fabric injected.

Spiderweb edges

Edges carry currentUtil as fabric-driven story. Tesla reverse damping from Chapter 9 modifies edge update — forward ease, reverse resist.

Tidewalker tier EnableSpiderwebGraph allows override — operator responsibility rises; Chapter 18 covenant applies.

simulatedChipCycles

Accumulated cycles are comparative telemetry — useful for session shape, not billing. Headless CI can track monotonic increase as dispatch health.

Mirror anatomy 14 — units, edges, and sysfs

Depth pass 14 on hardwareFabric literacy.

Per-core units[]

Each unit entry tracks operationalFreqMHz, utilization subfunctions, voltage, temperature, power proxy. Entry 13 is not a physical core map — it is dashboard metaphor aligned to sysfs when Puny AutoUseRealClocks reads real MHz.

Compare list Hardware output to grep STATUS — numbers should move together when fabric injected.

Spiderweb edges

Edges carry currentUtil as fabric-driven story. Tesla reverse damping from Chapter 9 modifies edge update — forward ease, reverse resist.

Tidewalker tier EnableSpiderwebGraph allows override — operator responsibility rises; Chapter 18 covenant applies.

simulatedChipCycles

Accumulated cycles are comparative telemetry — useful for session shape, not billing. Headless CI can track monotonic increase as dispatch health.

Mirror anatomy 15 — units, edges, and sysfs

Depth pass 15 on hardwareFabric literacy.

Per-core units[]

Each unit entry tracks operationalFreqMHz, utilization subfunctions, voltage, temperature, power proxy.

Entry 14 is not a physical core map — it is dashboard metaphor aligned to sysfs when Puny

AutoUseRealClocks reads real MHz.

Compare list Hardware output to grep STATUS — numbers should move together when fabric injected.

Spiderweb edges

Edges carry currentUtil as fabric-driven story. Tesla reverse damping from Chapter 9 modifies edge update

— forward ease, reverse resist.

Tidewalker tier EnableSpiderwebGraph allows override — operator responsibility rises; Chapter 18 covenant applies.

simulatedChipCycles

Accumulated cycles are comparative telemetry — useful for session shape, not billing. Headless CI can track monotonic increase as dispatch health.

Mirror anatomy 16 — units, edges, and sysfs

Depth pass 16 on hardwareFabric literacy.

Per-core units[]

Each unit entry tracks operationalFreqMHz, utilization subfunctions, voltage, temperature, power proxy.

Entry 15 is not a physical core map — it is dashboard metaphor aligned to sysfs when Puny

AutoUseRealClocks reads real MHz.

Compare list Hardware output to grep STATUS — numbers should move together when fabric injected.

Spiderweb edges

Edges carry currentUtil as fabric-driven story. Tesla reverse damping from Chapter 9 modifies edge update

— forward ease, reverse resist.

Tidewalker tier EnableSpiderwebGraph allows override — operator responsibility rises; Chapter 18 covenant applies.

simulatedChipCycles

Accumulated cycles are comparative telemetry — useful for session shape, not billing. Headless CI can track monotonic increase as dispatch health.

Mirror anatomy 17 — units, edges, and sysfs

Depth pass 17 on hardwareFabric literacy.

Per-core units[]

Each unit entry tracks operationalFreqMHz, utilization subfunctions, voltage, temperature, power proxy. Entry 16 is not a physical core map — it is dashboard metaphor aligned to sysfs when Puny AutoUseRealClocks reads real MHz.

Compare list Hardware output to grep STATUS — numbers should move together when fabric injected.

Spiderweb edges

Edges carry currentUtil as fabric-driven story. Tesla reverse damping from Chapter 9 modifies edge update — forward ease, reverse resist.

Tidewalker tier EnableSpiderwebGraph allows override — operator responsibility rises; Chapter 18 covenant applies.

simulatedChipCycles

Accumulated cycles are comparative telemetry — useful for session shape, not billing. Headless CI can track monotonic increase as dispatch health.

Mirror anatomy 18 — units, edges, and sysfs

Depth pass 18 on hardwareFabric literacy.

Per-core units[]

Each unit entry tracks operationalFreqMHz, utilization subfunctions, voltage, temperature, power proxy. Entry 17 is not a physical core map — it is dashboard metaphor aligned to sysfs when Puny AutoUseRealClocks reads real MHz.

Compare list Hardware output to grep STATUS — numbers should move together when fabric injected.

Spiderweb edges

Edges carry currentUtil as fabric-driven story. Tesla reverse damping from Chapter 9 modifies edge update — forward ease, reverse resist.

Tidewalker tier EnableSpiderwebGraph allows override — operator responsibility rises; Chapter 18 covenant applies.

simulatedChipCycles

Accumulated cycles are comparative telemetry — useful for session shape, not billing. Headless CI can track monotonic increase as dispatch health.

Mirror anatomy 19 — units, edges, and sysfs

Depth pass 19 on hardwareFabric literacy.

Per-core units[]

Each unit entry tracks operationalFreqMHz, utilization subfunctions, voltage, temperature, power proxy. Entry 18 is not a physical core map — it is dashboard metaphor aligned to sysfs when Puny AutoUseRealClocks reads real MHz.

Compare list Hardware output to grep STATUS — numbers should move together when fabric injected.

Spiderweb edges

Edges carry currentUtil as fabric-driven story. Tesla reverse damping from Chapter 9 modifies edge update — forward ease, reverse resist.

Tidewalker tier EnableSpiderwebGraph allows override — operator responsibility rises; Chapter 18 covenant applies.

simulatedChipCycles

Accumulated cycles are comparative telemetry — useful for session shape, not billing. Headless CI can track monotonic increase as dispatch health.

Mirror anatomy 20 — units, edges, and sysfs

Depth pass 20 on hardwareFabric literacy.

Per-core units[]

Each unit entry tracks operationalFreqMHz, utilization subfunctions, voltage, temperature, power proxy. Entry 19 is not a physical core map — it is dashboard metaphor aligned to sysfs when Puny AutoUseRealClocks reads real MHz.

Compare list Hardware output to grep STATUS — numbers should move together when fabric injected.

Spiderweb edges

Edges carry currentUtil as fabric-driven story. Tesla reverse damping from Chapter 9 modifies edge update — forward ease, reverse resist.

Tidewalker tier EnableSpiderwebGraph allows override — operator responsibility rises; Chapter 18 covenant applies.

simulatedChipCycles

Accumulated cycles are comparative telemetry — useful for session shape, not billing. Headless CI can track monotonic increase as dispatch health.

Mirror anatomy 21 — units, edges, and sysfs

Depth pass 21 on hardwareFabric literacy.

Per-core units[]

Each unit entry tracks operationalFreqMHz, utilization subfunctions, voltage, temperature, power proxy. Entry 20 is not a physical core map — it is dashboard metaphor aligned to sysfs when Puny AutoUseRealClocks reads real MHz.

Compare list Hardware output to grep STATUS — numbers should move together when fabric injected.

Spiderweb edges

Edges carry currentUtil as fabric-driven story. Tesla reverse damping from Chapter 9 modifies edge update — forward ease, reverse resist.

Tidewalker tier EnableSpiderwebGraph allows override — operator responsibility rises; Chapter 18 covenant applies.

simulatedChipCycles

Accumulated cycles are comparative telemetry — useful for session shape, not billing. Headless CI can track monotonic increase as dispatch health.

Mirror anatomy 22 — units, edges, and sysfs

Depth pass 22 on hardwareFabric literacy.

Per-core units[]

Each unit entry tracks operationalFreqMHz, utilization subfunctions, voltage, temperature, power proxy.
Entry 21 is not a physical core map — it is dashboard metaphor aligned to sysfs when Puny
AutoUseRealClocks reads real MHz.

Compare list Hardware output to grep STATUS — numbers should move together when fabric injected.

Spiderweb edges

Edges carry currentUtil as fabric-driven story. Tesla reverse damping from Chapter 9 modifies edge update
— forward ease, reverse resist.

Tidewalker tier EnableSpiderwebGraph allows override — operator responsibility rises; Chapter 18
covenant applies.

simulatedChipCycles

Accumulated cycles are comparative telemetry — useful for session shape, not billing. Headless CI can
track monotonic increase as dispatch health.

Mirror anatomy 23 — units, edges, and sysfs

Depth pass 23 on hardwareFabric literacy.

Per-core units[]

Each unit entry tracks operationalFreqMHz, utilization subfunctions, voltage, temperature, power proxy.
Entry 22 is not a physical core map — it is dashboard metaphor aligned to sysfs when Puny
AutoUseRealClocks reads real MHz.

Compare list Hardware output to grep STATUS — numbers should move together when fabric injected.

Spiderweb edges

Edges carry currentUtil as fabric-driven story. Tesla reverse damping from Chapter 9 modifies edge update
— forward ease, reverse resist.

Tidewalker tier EnableSpiderwebGraph allows override — operator responsibility rises; Chapter 18
covenant applies.

simulatedChipCycles

Accumulated cycles are comparative telemetry — useful for session shape, not billing. Headless CI can
track monotonic increase as dispatch health.

Mirror anatomy 24 — units, edges, and sysfs

Depth pass 24 on hardwareFabric literacy.

Per-core units[]

Each unit entry tracks operationalFreqMHz, utilization subfunctions, voltage, temperature, power proxy. Entry 23 is not a physical core map — it is dashboard metaphor aligned to sysfs when Puny AutoUseRealClocks reads real MHz.

Compare list Hardware output to grep STATUS — numbers should move together when fabric injected.

Spiderweb edges

Edges carry currentUtil as fabric-driven story. Tesla reverse damping from Chapter 9 modifies edge update — forward ease, reverse resist.

Tidewalker tier EnableSpiderwebGraph allows override — operator responsibility rises; Chapter 18 covenant applies.

simulatedChipCycles

Accumulated cycles are comparative telemetry — useful for session shape, not billing. Headless CI can track monotonic increase as dispatch health.

Mirror anatomy 25 — units, edges, and sysfs

Depth pass 25 on hardwareFabric literacy.

Per-core units[]

Each unit entry tracks operationalFreqMHz, utilization subfunctions, voltage, temperature, power proxy. Entry 24 is not a physical core map — it is dashboard metaphor aligned to sysfs when Puny AutoUseRealClocks reads real MHz.

Compare list Hardware output to grep STATUS — numbers should move together when fabric injected.

Spiderweb edges

Edges carry currentUtil as fabric-driven story. Tesla reverse damping from Chapter 9 modifies edge update — forward ease, reverse resist.

Tidewalker tier EnableSpiderwebGraph allows override — operator responsibility rises; Chapter 18 covenant applies.

simulatedChipCycles

Accumulated cycles are comparative telemetry — useful for session shape, not billing. Headless CI can track monotonic increase as dispatch health.

Mirror anatomy 26 — units, edges, and sysfs

Depth pass 26 on hardwareFabric literacy.

Per-core units[]

Each unit entry tracks operationalFreqMHz, utilization subfunctions, voltage, temperature, power proxy. Entry 25 is not a physical core map — it is dashboard metaphor aligned to sysfs when Puny AutoUseRealClocks reads real MHz.

Compare list Hardware output to grep STATUS — numbers should move together when fabric injected.

Spiderweb edges

Edges carry currentUtil as fabric-driven story. Tesla reverse damping from Chapter 9 modifies edge update — forward ease, reverse resist.

Tidewalker tier EnableSpiderwebGraph allows override — operator responsibility rises; Chapter 18 covenant applies.

simulatedChipCycles

Accumulated cycles are comparative telemetry — useful for session shape, not billing. Headless CI can track monotonic increase as dispatch health.

Mirror anatomy 27 — units, edges, and sysfs

Depth pass 27 on hardwareFabric literacy.

Per-core units[]

Each unit entry tracks operationalFreqMHz, utilization subfunctions, voltage, temperature, power proxy. Entry 26 is not a physical core map — it is dashboard metaphor aligned to sysfs when Puny AutoUseRealClocks reads real MHz.

Compare list Hardware output to grep STATUS — numbers should move together when fabric injected.

Spiderweb edges

Edges carry currentUtil as fabric-driven story. Tesla reverse damping from Chapter 9 modifies edge update — forward ease, reverse resist.

Tidewalker tier EnableSpiderwebGraph allows override — operator responsibility rises; Chapter 18 covenant applies.

simulatedChipCycles

Accumulated cycles are comparative telemetry — useful for session shape, not billing. Headless CI can track monotonic increase as dispatch health.

Mirror anatomy 28 — units, edges, and sysfs

Depth pass 28 on hardwareFabric literacy.

Per-core units[]

Each unit entry tracks operationalFreqMHz, utilization subfunctions, voltage, temperature, power proxy. Entry 27 is not a physical core map — it is dashboard metaphor aligned to sysfs when Puny AutoUseRealClocks reads real MHz.

Compare list Hardware output to grep STATUS — numbers should move together when fabric injected.

Spiderweb edges

Edges carry currentUtil as fabric-driven story. Tesla reverse damping from Chapter 9 modifies edge update — forward ease, reverse resist.

Tidewalker tier EnableSpiderwebGraph allows override — operator responsibility rises; Chapter 18 covenant applies.

simulatedChipCycles

Accumulated cycles are comparative telemetry — useful for session shape, not billing. Headless CI can track monotonic increase as dispatch health.

Mirror anatomy 29 — units, edges, and sysfs

Depth pass 29 on hardwareFabric literacy.

Per-core units[]

Each unit entry tracks operationalFreqMHz, utilization subfunctions, voltage, temperature, power proxy. Entry 28 is not a physical core map — it is dashboard metaphor aligned to sysfs when Puny

AutoUseRealClocks reads real MHz.

Compare list Hardware output to grep STATUS — numbers should move together when fabric injected.

Spiderweb edges

Edges carry currentUtil as fabric-driven story. Tesla reverse damping from Chapter 9 modifies edge update — forward ease, reverse resist.

Tidewalker tier EnableSpiderwebGraph allows override — operator responsibility rises; Chapter 18 covenant applies.

simulatedChipCycles

Accumulated cycles are comparative telemetry — useful for session shape, not billing. Headless CI can track monotonic increase as dispatch health.

Mirror anatomy 30 — units, edges, and sysfs

Depth pass 30 on hardwareFabric literacy.

Per-core units[]

Each unit entry tracks operationalFreqMHz, utilization subfunctions, voltage, temperature, power proxy. Entry 29 is not a physical core map — it is dashboard metaphor aligned to sysfs when Puny AutoUseRealClocks reads real MHz.

Compare list Hardware output to grep STATUS — numbers should move together when fabric injected.

Spiderweb edges

Edges carry currentUtil as fabric-driven story. Tesla reverse damping from Chapter 9 modifies edge update — forward ease, reverse resist.

Tidewalker tier EnableSpiderwebGraph allows override — operator responsibility rises; Chapter 18 covenant applies.

simulatedChipCycles

Accumulated cycles are comparative telemetry — useful for session shape, not billing. Headless CI can track monotonic increase as dispatch health.

Mirror anatomy 31 — units, edges, and sysfs

Depth pass 31 on hardwareFabric literacy.

Per-core units[]

Each unit entry tracks operationalFreqMHz, utilization subfunctions, voltage, temperature, power proxy.

Entry 30 is not a physical core map — it is dashboard metaphor aligned to sysfs when Puny

AutoUseRealClocks reads real MHz.

Compare list Hardware output to grep STATUS — numbers should move together when fabric injected.

Spiderweb edges

Edges carry currentUtil as fabric-driven story. Tesla reverse damping from Chapter 9 modifies edge update — forward ease, reverse resist.

Tidewalker tier EnableSpiderwebGraph allows override — operator responsibility rises; Chapter 18 covenant applies.

simulatedChipCycles

Accumulated cycles are comparative telemetry — useful for session shape, not billing. Headless CI can track monotonic increase as dispatch health.

Mirror anatomy 32 — units, edges, and sysfs

Depth pass 32 on hardwareFabric literacy.

Per-core units[]

Each unit entry tracks operationalFreqMHz, utilization subfunctions, voltage, temperature, power proxy.

Entry 31 is not a physical core map — it is dashboard metaphor aligned to sysfs when Puny

AutoUseRealClocks reads real MHz.

Compare list Hardware output to grep STATUS — numbers should move together when fabric injected.

Spiderweb edges

Edges carry currentUtil as fabric-driven story. Tesla reverse damping from Chapter 9 modifies edge update — forward ease, reverse resist.

Tidewalker tier EnableSpiderwebGraph allows override — operator responsibility rises; Chapter 18 covenant applies.

simulatedChipCycles

Accumulated cycles are comparative telemetry — useful for session shape, not billing. Headless CI can track monotonic increase as dispatch health.

Mirror anatomy 33 — units, edges, and sysfs

Depth pass 33 on hardwareFabric literacy.

Per-core units[]

Each unit entry tracks operationalFreqMHz, utilization subfunctions, voltage, temperature, power proxy. Entry 32 is not a physical core map — it is dashboard metaphor aligned to sysfs when Puny AutoUseRealClocks reads real MHz.

Compare list Hardware output to grep STATUS — numbers should move together when fabric injected.

Spiderweb edges

Edges carry currentUtil as fabric-driven story. Tesla reverse damping from Chapter 9 modifies edge update — forward ease, reverse resist.

Tidewalker tier EnableSpiderwebGraph allows override — operator responsibility rises; Chapter 18 covenant applies.

simulatedChipCycles

Accumulated cycles are comparative telemetry — useful for session shape, not billing. Headless CI can track monotonic increase as dispatch health.

Mirror anatomy 34 — units, edges, and sysfs

Depth pass 34 on hardwareFabric literacy.

Per-core units[]

Each unit entry tracks operationalFreqMHz, utilization subfunctions, voltage, temperature, power proxy. Entry 33 is not a physical core map — it is dashboard metaphor aligned to sysfs when Puny AutoUseRealClocks reads real MHz.

Compare list Hardware output to grep STATUS — numbers should move together when fabric injected.

Spiderweb edges

Edges carry currentUtil as fabric-driven story. Tesla reverse damping from Chapter 9 modifies edge update — forward ease, reverse resist.

Tidewalker tier EnableSpiderwebGraph allows override — operator responsibility rises; Chapter 18 covenant applies.

simulatedChipCycles

Accumulated cycles are comparative telemetry — useful for session shape, not billing. Headless CI can track monotonic increase as dispatch health.

Mirror anatomy 35 — units, edges, and sysfs

Depth pass 35 on hardwareFabric literacy.

Per-core units[]

Each unit entry tracks operationalFreqMHz, utilization subfunctions, voltage, temperature, power proxy. Entry 34 is not a physical core map — it is dashboard metaphor aligned to sysfs when Puny AutoUseRealClocks reads real MHz.

Compare list Hardware output to grep STATUS — numbers should move together when fabric injected.

Spiderweb edges

Edges carry currentUtil as fabric-driven story. Tesla reverse damping from Chapter 9 modifies edge update — forward ease, reverse resist.

Tidewalker tier EnableSpiderwebGraph allows override — operator responsibility rises; Chapter 18 covenant applies.

simulatedChipCycles

Accumulated cycles are comparative telemetry — useful for session shape, not billing. Headless CI can track monotonic increase as dispatch health.

Mirror anatomy 36 — units, edges, and sysfs

Depth pass 36 on hardwareFabric literacy.

Per-core units[]

Each unit entry tracks operationalFreqMHz, utilization subfunctions, voltage, temperature, power proxy. Entry 35 is not a physical core map — it is dashboard metaphor aligned to sysfs when Puny AutoUseRealClocks reads real MHz.

Compare list Hardware output to grep STATUS — numbers should move together when fabric injected.

Spiderweb edges

Edges carry currentUtil as fabric-driven story. Tesla reverse damping from Chapter 9 modifies edge update — forward ease, reverse resist.

Tidewalker tier EnableSpiderwebGraph allows override — operator responsibility rises; Chapter 18 covenant applies.

simulatedChipCycles

Accumulated cycles are comparative telemetry — useful for session shape, not billing. Headless CI can track monotonic increase as dispatch health.

Evidence anchor — grep and sources

Major claims in this chapter anchored for reproducibility. Implemented = grep today; Metaphor = intuition; Philosophy = discipline.

Claim	Statement	Label	Evidence
hardwareFabric	Spiderweb graph mirror	Implemented	updateHardwareFromAnalogFields(
Fabric averages	Dashboard compression	Metaphor	Lossy – not SEM
Adept tier	SimulateSubMicron	Metaphor	Procedural detail – not lab SEM
list Hardware	Operator readout	Implemented	Prompt terminal command

```
util(core) ← sample(avg(Phi, Thermo, Flow))
```

```
grep -n 'hardwareFabric' Navigator/engine/
```

Source paths

- Navigator/engine/Spiderweb.cpp
- Navigator/engine/rtx.cpp

Chapter summary — before you turn the page

Spiderweb mirrors fabric averages read-only — dashboard compression, not SEM truth. Recover detail in stderr and Classic canvas when averages lie by loss.

Reading the Battlefield

CHAPTER 11 · OBSERVABILITY — READING THE BATTLEFIELD

Learning objectives

- Use ELLIE categories and STATUS block fields as primary witnesses.
- Operate prompt `set/list` for AnalogFields and Hardware.
- Enable RTXProbe with zero cost when off.
- Navigate NEXUS panel `https://127.0.0.1:9477/` and RTX Zero mode.
- Execute week-one lab correlating stderr, die bus, and packet field jsonl.

IN THIS CHAPTER

- | | |
|--|---|
| 1. Introduction — grep is forensic defense | 8. Chapter summary |
| 2. ELLIE logging categories | 9. Study questions |
| 3. Prompt terminal — partial but real | 10. ELLIE STATUS block — field by field |
| 4. RTXProbe | 11. Correlating three scales at panel |
| 5. NEXUS panel | 12. RTX Zero idle posture |
| 6. Week-one operator lab | 13. Observability drill block 1 |
| 7. Sovereign time preview | 14. Observability drill block 2 |

On the way — what you will learn

Observability is grep discipline made weekly habit. On the way you will partition ELLIE categories, read STATUS and THERMO cadence, correlate panel JSON with stderr, and build an operator journal that survives long sessions.

On the way — Observability stack

- MAIN, VULKAN, CANVAS, THERMO, STATUS
- RTX_PROBE and headless dispatch receipts
- Battlefield reading without screenshots alone
- Week-twelve drift detection

Introduction — grep is forensic defense

Trust stderr before screenshots. Time is linear — logs are timeline. Chapter 11 unifies ELLIE logging, prompt terminal, RTXProbe, NEXUS panel into one observability doctrine spanning AMOURANTHRTX and NEXUS-Shield.

Figure 11.1 — Logs, probes, panel — one battlefield.

ELLIE logging categories

MAIN, VULKAN, CANVAS, THERMO, STATUS, RTXPROBE. STATUS ~5s: FPS, GPU ms, VRAM, adaptive scale, entropy, boundary thermo, maintenance cost.

Figure 11.2 — SQUIDGIE and THERMO share stderr timelines — grep both.

```
grep -E '^(MAIN|VULKAN|CANVAS|THERMO|STATUS)' run.log
```

Prompt terminal — partial but real

```
set AnalogFields.GateFidelity 0.85
list Hardware
guide
```

set/list AnalogFields + Hardware.

Glassmorphism sliders / ImGui ESC — feasibility doc only, not hidden as shipped.

RTXProbe

RTX_PROBES=1 → GPU timestamps, invocation counts. Zero cost when off.

NEXUS panel

<https://127.0.0.1:9477/> — command, packets, threats, signals, DNS, library, system. RTX Zero ?rtx=1 — Aqua chrome, cache-first refresh.

Figure 11.4 — Panel slice beside packet perimeter — one battlefield map.

Week-one operator lab

1. ./linux.sh run or ./nexus.sh
2. Read STATUS/THERMO 60s
3. Mouse on classic — entropyThisFrame
4. Archive gatekeeper decision at panel

Sovereign time preview

Run your timeserver, verify at receive, grep SQUIDGIE — Chapter 19

Figure 11.3 — field jsonl timestamps cross-check sealed time preview.

Claim: NEXUS packet field — local jsonl perimeter

Chapter summary

Observability is weapon. Prior: Ch 10. Next: Ch 12.

Study questions

1. Six ELLIE categories?
2. STATUS fields?
3. RTXProbe enable?

4. RTX Zero?

5. Panel port?

ELLIE STATUS block — field by field

FPS — presentation cadence. GPU ms — dispatch cost witness. VRAM — budget Chapter 7. Adaptive scale — CFL interaction Chapter 9. Entropy — ThermoAccountant proxy. Boundary thermo — holographic boundary metaphor with number. Maintenance — prevMaintCost story.

Correlating three scales at panel

stderr THERMO + data_bus concept + NEXUS jsonl row + sealed time — Chapter 2 integration. Panel :9477 is human correlate surface, not automatic merge engine.

RTX Zero idle posture

?rtx=1 Aqua chrome cache-first — zero-cost idle when honored. Observability includes knowing when refresh paused vs threat stale.

Observability drill block 1

Structured grep and panel exercise 1 for muscle memory.

stderr rhythm

Session 1: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Observability drill block 2

Structured grep and panel exercise 2 for muscle memory.

stderr rhythm

Session 2: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Observability drill block 3

Structured grep and panel exercise 3 for muscle memory.

stderr rhythm

Session 3: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Observability drill block 4

Structured grep and panel exercise 4 for muscle memory.

stderr rhythm

Session 4: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Observability drill block 5

Structured grep and panel exercise 5 for muscle memory.

stderr rhythm

Session 5: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Observability drill block 6

Structured grep and panel exercise 6 for muscle memory.

stderr rhythm

Session 6: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Observability drill block 7

Structured grep and panel exercise 7 for muscle memory.

stderr rhythm

Session 7: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Observability drill block 8

Structured grep and panel exercise 8 for muscle memory.

stderr rhythm

Session 8: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Observability drill block 9

Structured grep and panel exercise 9 for muscle memory.

stderr rhythm

Session 9: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Observability drill block 10

Structured grep and panel exercise 10 for muscle memory.

stderr rhythm

Session 10: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Observability drill block 11

Structured grep and panel exercise 11 for muscle memory.

stderr rhythm

Session 11: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Observability drill block 12

Structured grep and panel exercise 12 for muscle memory.

stderr rhythm

Session 12: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Observability drill block 13

Structured grep and panel exercise 13 for muscle memory.

stderr rhythm

Session 13: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Observability drill block 14

Structured grep and panel exercise 14 for muscle memory.

stderr rhythm

Session 14: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Observability drill block 15

Structured grep and panel exercise 15 for muscle memory.

stderr rhythm

Session 15: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Observability drill block 16

Structured grep and panel exercise 16 for muscle memory.

stderr rhythm

Session 16: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Observability drill block 17

Structured grep and panel exercise 17 for muscle memory.

stderr rhythm

Session 17: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Observability drill block 18

Structured grep and panel exercise 18 for muscle memory.

stderr rhythm

Session 18: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Observability drill block 19

Structured grep and panel exercise 19 for muscle memory.

stderr rhythm

Session 19: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Observability drill block 20

Structured grep and panel exercise 20 for muscle memory.

stderr rhythm

Session 20: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Observability drill block 21

Structured grep and panel exercise 21 for muscle memory.

stderr rhythm

Session 21: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Observability drill block 22

Structured grep and panel exercise 22 for muscle memory.

stderr rhythm

Session 22: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Observability drill block 23

Structured grep and panel exercise 23 for muscle memory.

stderr rhythm

Session 23: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Observability drill block 24

Structured grep and panel exercise 24 for muscle memory.

stderr rhythm

Session 24: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Observability drill block 25

Structured grep and panel exercise 25 for muscle memory.

stderr rhythm

Session 25: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Observability drill block 26

Structured grep and panel exercise 26 for muscle memory.

stderr rhythm

Session 26: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Observability drill block 27

Structured grep and panel exercise 27 for muscle memory.

stderr rhythm

Session 27: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Observability drill block 28

Structured grep and panel exercise 28 for muscle memory.

stderr rhythm

Session 28: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Observability drill block 29

Structured grep and panel exercise 29 for muscle memory.

stderr rhythm

Session 29: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Observability drill block 30

Structured grep and panel exercise 30 for muscle memory.

stderr rhythm

Session 30: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Observability drill block 31

Structured grep and panel exercise 31 for muscle memory.

stderr rhythm

Session 31: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Observability drill block 32

Structured grep and panel exercise 32 for muscle memory.

stderr rhythm

Session 32: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Observability drill block 33

Structured grep and panel exercise 33 for muscle memory.

stderr rhythm

Session 33: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Observability drill block 34

Structured grep and panel exercise 34 for muscle memory.

stderr rhythm

Session 34: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Observability drill block 35

Structured grep and panel exercise 35 for muscle memory.

stderr rhythm

Session 35: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Observability drill block 36

Structured grep and panel exercise 36 for muscle memory.

stderr rhythm

Session 36: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Observability drill block 37

Structured grep and panel exercise 37 for muscle memory.

stderr rhythm

Session 37: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Observability drill block 38

Structured grep and panel exercise 38 for muscle memory.

stderr rhythm

Session 38: tee run.log for 120 seconds. Count STATUS lines — expect ~24 at 5s cadence. Note entropy and boundary thermo drift when injecting mouse on fabric.

Correlate THERMO spike with dispatch steps in data_bus mirror conceptually — slot 28 should rise monotonically.

panel archive

Open NEXUS panel loopback. Archive one gatekeeper verdict jsonl row — USER_OK or SUSPICIOUS. Redact nothing; local-first truth.

RTX Zero mode ?rtx=1: confirm cache-first refresh does not hide stale threat panel — note behavior for operator journal.

cross-product correlation

Timestamp sealed time from FieldSocket concept against jsonl row time — Chapter 19 preview. If disagree, flag SQUIDGIE research path.

Evidence anchor — grep and sources

Major claims in this chapter anchored for reproducibility. Implemented = grep today; Metaphor
= intuition; Philosophy = discipline.

Claim	Statement	Label	Evidence
ELLIE categories	Categorized stderr	Implemented	MAIN/VULKAN/CANVAS/THERMO/STATUS
Panel :9477	Human correlation UI	Implemented	NEXUS local panel
Screenshot culture	Pretty frames as proof	Philosophy	Rejected – grep first
Operator journal	Weekly rhythm	Philosophy	Receipts over vibes

```
observability = stderr_categories × panel_correlation × jsonl_archive
```

```
grep -E '^(MAIN|VULKAN|THERMO)' run.log | tail -40
```

Source paths

- Navigator/engine/ELLIE.hpp
- NEXUS-Shield/panel_server.py

Chapter summary — before you turn the page

Observability is categorized stderr, panel correlation, and operator journal discipline — receipts culture over screenshot culture.

The Rocks We Do Not Hide

CHAPTER 12 · REALITY VS THEORY — THE ROCKS

Learning objectives

- Apply status tags (impl, meta, phil, vis) to any stack claim.
- Recite the complete honesty table from memory with operator reality column.
- Resolve disputes using trust ordering: headers, stderr, default run path.
- Separate product licenses and roles without merger mythology.
- Use capstone equations at correct measurement layer.

IN THIS CHAPTER

- | | |
|---|--|
| 1. Introduction — bookmark this chapter | 9. Status tags — complete operator legend |
| 2. The honesty table — complete | 10. Engineering core chapters 2–11 — what was real |
| 3. What to trust | 11. Enjoy the field — honestly |
| 4. Product boundaries | 12. Honesty application 1 — dispute resolution |
| 5. Capstone equations | 13. Honesty application 2 — dispute resolution |
| 6. Category errors catalog | 14. Honesty application 3 — dispute resolution |
| 7. Chapter summary | |
| 8. Study questions | |

On the way — what you will learn

This is the chapter you bookmark. On the way you will catalog every **rock** — what is implemented, metaphor, philosophy, visual — and practice refusing vendor cosmology, joule fantasy, and shader-as-spectrum lies.

On the way — Honesty mirror

- Full honesty table with stack examples
- Category error encyclopedia
- When poetry is welcome versus forbidden
- Return here when marketing sells 3D hype

Introduction — bookmark this chapter

Field Technology v5 is serious because it labels poetry before newcomers confuse it with measurement. When someone sells cosmology, return here. Theory inspires vocabulary; implementation is what you grep, set, screenshot — with labels.

Figure 12.1 — Metaphor glow vs measured grid.

The honesty table — complete

Figure 12.2 — Layers share words, not measurements — four labels prevent summing.

Claim (theory / marketing)	Operator reality	Label
Greatest weapon	Field literacy + local tools	Philosophy
Living thermodynamic computer	ThermoAccountant in logs	Implemented
Packet field sees everything	Local sockets + heuristics	Implemented
RF planetary shell	planetary_weave.comp visual	Visual
Landauer joules from GPU	Proxy integral entropyThisFrame	Metaphor
Sub-micron SEM	Procedural pixel detail	Metaphor
Love / God chapters	Sacred operator language	Philosophy
Queen holds all gates	QUEEN_READY when built	Implemented
KILROY kernel on host today	QEMU/bare-metal image; host may run generic Linux	Implemented image
Propalactic cosmic weave	PropalacticScale knob on Phi	Metaphor
Tamper abort / sealed time	TotalTime::seal() + verify posture	Implemented session
Hardware-gate fidelity	GateFidelity sharpens Flow; spiderweb util mirror	Implemented
Tesla valve in chassis	TESLA_R_* constants; data_bus 31/34	Metaphor
96% lies device cannot drift	Security posture narrative	Philosophy
data_bus is SI telemetry	64-word packed dashboard per dispatch	Implemented
x86.comp is decorative	Default Field Die product path	Implemented
FIELD_LAYOUT_VERSION cosmetic	Host/shader layout contract v5	Implemented

Claim (theory / marketing)	Operator reality	Label
ELLIE is optional polish	Primary observability spine	Implemented

What to trust

1. Headers over cover art
2. stderr over screenshots for thermo numbers
3. ./linux.sh run default = Field Die
4. Headless dispatch — valid CI signal

Product boundaries

Product	License	Role
AMOURANTHRTX	GPL v3 or commercial	Vulkan Field Die
NEXUS-Shield	MIT	Endpoint security
Queen	Field sovereign browser	RTX + gates in-engine
Field Primer	CC BY-NC-SA 4.0	This textbook

Capstone equations

Domain	Relation	Layer
Wave CFL	$c \cdot \Delta t / \Delta x \leq 1$	Host guard
Diffusion CFL	$\alpha \cdot \Delta t / \Delta x^2 \leq 1$	Host guard
Entropy proxy	$\dot{S} \approx \text{fieldWork} + \text{probeDiss} + \text{prevMaint}$	ThermoAccountant
Shannon (NEXUS)	$H = -\sum p_i \log_2 p_i$	Entropy Oracle files
Landauer (theory)	$E_{\min} = k_B T \ln 2$	Theory — proxy labeled

Category errors catalog

Do not treat shader art as instrumentation. **Do not** treat jsonl verdict as joules. **Do not** treat guest RAM metaphor as cloud perimeter. **Do not** conflate Shannon file H with ThermoAccountant entropy.

Chapter summary

Enjoy the field — honestly. Next: Chapter 13 Landauer Prior engineering core: Chapters 2–11.

Figure 12.4 — Category error: Visual ionosphere $\neq$ instrumentation.

Study questions

1. Five honesty rows from memory.
2. stderr vs screenshots?
3. MIT product?
4. Diffusion CFL?
5. PropalacticScale label?

Status tags — complete operator legend

Tag	Meaning
Implemented	Grep, set, screenshot in source
Metaphor	Poetic naming; not SI units
Philosophy	Operator discipline
Visual	Shader art; not instrumentation

Engineering core chapters 2–11 — what was real

Ch2 three scales. Ch3 thermo fabric. Ch4 entropy layers. Ch5 packet field. Ch6 RF separation. Ch7 dispatch. Ch8 die bus. Ch9 CFL Tesla. Ch10 spiderweb mirror. Ch11 observability. All labeled. Ch13+ creditors and sacred long-form follow with same posture.

Figure 12.3 — Seductive details weed: signaled figures only (Mayer coherence).

Enjoy the field — honestly

Theory inspires vocabulary. Implementation is grep, set, screenshot. We do not hide the rocks.

Honesty application 1 — dispute resolution

Worked example 1 applying rocks table.

Dispute setup

Example 1: A vendor claims sub-micron SEM fidelity from spiderweb screenshot. Operator opens Chapter 12 table — procedural pixel detail, Metaphor label. Resolution: screenshot is Visual pedagogy, not lab.

Second dispute: jsonl shows SUSPICIOUS — colleague says GPU is compromised. Operator separates packet field local sockets from Vulkan dispatch — corroborate before permanent KILL.

Trust ordering

grep FieldRtxFieldAbs.hpp beats marketing PDF. run.log THERMO beats influencer stream. Default x86 die beats rumor about decorative engine.

Capstone equation layer

Landauer $E_{\min}$ is theory anchor — entropyThisFrame remains proxy. Shannon H on executable is NEXUS — not mixed into thermo boundary without label.

Honesty application 2 — dispute resolution

Worked example 2 applying rocks table.

Dispute setup

Example 2: A vendor claims sub-micron SEM fidelity from spiderweb screenshot. Operator opens Chapter 12 table — procedural pixel detail, Metaphor label. Resolution: screenshot is Visual pedagogy, not lab.

Second dispute: jsonl shows SUSPICIOUS — colleague says GPU is compromised. Operator separates packet field local sockets from Vulkan dispatch — corroborate before permanent KILL.

Trust ordering

grep FieldRtxFieldAbs.hpp beats marketing PDF. run.log THERMO beats influencer stream. Default x86 die beats rumor about decorative engine.

Capstone equation layer

Landauer E_min is theory anchor — entropyThisFrame remains proxy. Shannon H on executable is NEXUS — not mixed into thermo boundary without label.

Honesty application 3 — dispute resolution

Worked example 3 applying rocks table.

Dispute setup

Example 3: A vendor claims sub-micron SEM fidelity from spiderweb screenshot. Operator opens Chapter 12 table — procedural pixel detail, Metaphor label. Resolution: screenshot is Visual pedagogy, not lab.

Second dispute: jsonl shows SUSPICIOUS — colleague says GPU is compromised. Operator separates packet field local sockets from Vulkan dispatch — corroborate before permanent KILL.

Trust ordering

grep FieldRtxFieldAbs.hpp beats marketing PDF. run.log THERMO beats influencer stream. Default x86 die beats rumor about decorative engine.

Capstone equation layer

Landauer E_min is theory anchor — entropyThisFrame remains proxy. Shannon H on executable is NEXUS — not mixed into thermo boundary without label.

Honesty application 4 — dispute resolution

Worked example 4 applying rocks table.

Dispute setup

Example 4: A vendor claims sub-micron SEM fidelity from spiderweb screenshot. Operator opens Chapter 12 table — procedural pixel detail, Metaphor label. Resolution: screenshot is Visual pedagogy, not lab.

Second dispute: jsonl shows SUSPICIOUS — colleague says GPU is compromised. Operator separates packet field local sockets from Vulkan dispatch — corroborate before permanent KILL.

Trust ordering

grep FieldRtxFieldAbs.hpp beats marketing PDF. run.log THERMO beats influencer stream. Default x86 die beats rumor about decorative engine.

Capstone equation layer

Landauer E_min is theory anchor — entropyThisFrame remains proxy. Shannon H on executable is NEXUS — not mixed into thermo boundary without label.

Honesty application 5 — dispute resolution

Worked example 5 applying rocks table.

Dispute setup

Example 5: A vendor claims sub-micron SEM fidelity from spiderweb screenshot. Operator opens Chapter 12 table — procedural pixel detail, Metaphor label. Resolution: screenshot is Visual pedagogy, not lab.

Second dispute: jsonl shows SUSPICIOUS — colleague says GPU is compromised. Operator separates packet field local sockets from Vulkan dispatch — corroborate before permanent KILL.

Trust ordering

grep FieldRtxFieldAbs.hpp beats marketing PDF. run.log THERMO beats influencer stream. Default x86 die beats rumor about decorative engine.

Capstone equation layer

Landauer E_min is theory anchor — entropyThisFrame remains proxy. Shannon H on executable is NEXUS — not mixed into thermo boundary without label.

Honesty application 6 — dispute resolution

Worked example 6 applying rocks table.

Dispute setup

Example 6: A vendor claims sub-micron SEM fidelity from spiderweb screenshot. Operator opens Chapter 12 table — procedural pixel detail, Metaphor label. Resolution: screenshot is Visual pedagogy, not lab.

Second dispute: jsonl shows SUSPICIOUS — colleague says GPU is compromised. Operator separates packet field local sockets from Vulkan dispatch — corroborate before permanent KILL.

Trust ordering

grep FieldRtxFieldAbs.hpp beats marketing PDF. run.log THERMO beats influencer stream. Default x86 die beats rumor about decorative engine.

Capstone equation layer

Landauer E_min is theory anchor — entropyThisFrame remains proxy. Shannon H on executable is NEXUS — not mixed into thermo boundary without label.

Honesty application 7 — dispute resolution

Worked example 7 applying rocks table.

Dispute setup

Example 7: A vendor claims sub-micron SEM fidelity from spiderweb screenshot. Operator opens Chapter 12 table — procedural pixel detail, Metaphor label. Resolution: screenshot is Visual pedagogy, not lab.

Second dispute: jsonl shows SUSPICIOUS — colleague says GPU is compromised. Operator separates packet field local sockets from Vulkan dispatch — corroborate before permanent KILL.

Trust ordering

grep FieldRtxFieldAbs.hpp beats marketing PDF. run.log THERMO beats influencer stream. Default x86 die beats rumor about decorative engine.

Capstone equation layer

Landauer E_min is theory anchor — entropyThisFrame remains proxy. Shannon H on executable is NEXUS — not mixed into thermo boundary without label.

Honesty application 8 — dispute resolution

Worked example 8 applying rocks table.

Dispute setup

Example 8: A vendor claims sub-micron SEM fidelity from spiderweb screenshot. Operator opens Chapter 12 table — procedural pixel detail, Metaphor label. Resolution: screenshot is Visual pedagogy, not lab.

Second dispute: jsonl shows SUSPICIOUS — colleague says GPU is compromised. Operator separates packet field local sockets from Vulkan dispatch — corroborate before permanent KILL.

Trust ordering

grep FieldRtxFieldAbs.hpp beats marketing PDF. run.log THERMO beats influencer stream. Default x86 die beats rumor about decorative engine.

Capstone equation layer

Landauer E_min is theory anchor — entropyThisFrame remains proxy. Shannon H on executable is NEXUS — not mixed into thermo boundary without label.

Honesty application 9 — dispute resolution

Worked example 9 applying rocks table.

Dispute setup

Example 9: A vendor claims sub-micron SEM fidelity from spiderweb screenshot. Operator opens Chapter 12 table — procedural pixel detail, Metaphor label. Resolution: screenshot is Visual pedagogy, not lab.

Second dispute: jsonl shows SUSPICIOUS — colleague says GPU is compromised. Operator separates packet field local sockets from Vulkan dispatch — corroborate before permanent KILL.

Trust ordering

grep FieldRtxFieldAbs.hpp beats marketing PDF. run.log THERMO beats influencer stream. Default x86 die beats rumor about decorative engine.

Capstone equation layer

Landauer E_min is theory anchor — entropyThisFrame remains proxy. Shannon H on executable is NEXUS — not mixed into thermo boundary without label.

Honesty application 10 — dispute resolution

Worked example 10 applying rocks table.

Dispute setup

Example 10: A vendor claims sub-micron SEM fidelity from spiderweb screenshot. Operator opens Chapter 12 table — procedural pixel detail, Metaphor label. Resolution: screenshot is Visual pedagogy, not lab.

Second dispute: jsonl shows SUSPICIOUS — colleague says GPU is compromised. Operator separates packet field local sockets from Vulkan dispatch — corroborate before permanent KILL.

Trust ordering

grep FieldRtxFieldAbs.hpp beats marketing PDF. run.log THERMO beats influencer stream. Default x86 die beats rumor about decorative engine.

Capstone equation layer

Landauer E_min is theory anchor — entropyThisFrame remains proxy. Shannon H on executable is NEXUS — not mixed into thermo boundary without label.

Honesty application 11 — dispute resolution

Worked example 11 applying rocks table.

Dispute setup

Example 11: A vendor claims sub-micron SEM fidelity from spiderweb screenshot. Operator opens Chapter 12 table — procedural pixel detail, Metaphor label. Resolution: screenshot is Visual pedagogy, not lab.

Second dispute: jsonl shows SUSPICIOUS — colleague says GPU is compromised. Operator separates packet field local sockets from Vulkan dispatch — corroborate before permanent KILL.

Trust ordering

grep FieldRtxFieldAbs.hpp beats marketing PDF. run.log THERMO beats influencer stream. Default x86 die beats rumor about decorative engine.

Capstone equation layer

Landauer E_min is theory anchor — entropyThisFrame remains proxy. Shannon H on executable is NEXUS — not mixed into thermo boundary without label.

Honesty application 12 — dispute resolution

Worked example 12 applying rocks table.

Dispute setup

Example 12: A vendor claims sub-micron SEM fidelity from spiderweb screenshot. Operator opens Chapter 12 table — procedural pixel detail, Metaphor label. Resolution: screenshot is Visual pedagogy, not lab.

Second dispute: jsonl shows SUSPICIOUS — colleague says GPU is compromised. Operator separates packet field local sockets from Vulkan dispatch — corroborate before permanent KILL.

Trust ordering

grep FieldRtxFieldAbs.hpp beats marketing PDF. run.log THERMO beats influencer stream. Default x86 die beats rumor about decorative engine.

Capstone equation layer

Landauer E_min is theory anchor — entropyThisFrame remains proxy. Shannon H on executable is NEXUS — not mixed into thermo boundary without label.

Honesty application 13 — dispute resolution

Worked example 13 applying rocks table.

Dispute setup

Example 13: A vendor claims sub-micron SEM fidelity from spiderweb screenshot. Operator opens Chapter 12 table — procedural pixel detail, Metaphor label. Resolution: screenshot is Visual pedagogy, not lab.

Second dispute: jsonl shows SUSPICIOUS — colleague says GPU is compromised. Operator separates packet field local sockets from Vulkan dispatch — corroborate before permanent KILL.

Trust ordering

grep FieldRtxFieldAbs.hpp beats marketing PDF. run.log THERMO beats influencer stream. Default x86 die beats rumor about decorative engine.

Capstone equation layer

Landauer E_min is theory anchor — entropyThisFrame remains proxy. Shannon H on executable is NEXUS — not mixed into thermo boundary without label.

Honesty application 14 — dispute resolution

Worked example 14 applying rocks table.

Dispute setup

Example 14: A vendor claims sub-micron SEM fidelity from spiderweb screenshot. Operator opens Chapter 12 table — procedural pixel detail, Metaphor label. Resolution: screenshot is Visual pedagogy, not lab.

Second dispute: jsonl shows SUSPICIOUS — colleague says GPU is compromised. Operator separates packet field local sockets from Vulkan dispatch — corroborate before permanent KILL.

Trust ordering

grep FieldRtxFieldAbs.hpp beats marketing PDF. run.log THERMO beats influencer stream. Default x86 die beats rumor about decorative engine.

Capstone equation layer

Landauer E_min is theory anchor — entropyThisFrame remains proxy. Shannon H on executable is NEXUS — not mixed into thermo boundary without label.

Honesty application 15 — dispute resolution

Worked example 15 applying rocks table.

Dispute setup

Example 15: A vendor claims sub-micron SEM fidelity from spiderweb screenshot. Operator opens Chapter 12 table — procedural pixel detail, Metaphor label. Resolution: screenshot is Visual pedagogy, not lab.

Second dispute: jsonl shows SUSPICIOUS — colleague says GPU is compromised. Operator separates packet field local sockets from Vulkan dispatch — corroborate before permanent KILL.

Trust ordering

grep FieldRtxFieldAbs.hpp beats marketing PDF. run.log THERMO beats influencer stream. Default x86 die beats rumor about decorative engine.

Capstone equation layer

Landauer E_min is theory anchor — entropyThisFrame remains proxy. Shannon H on executable is NEXUS — not mixed into thermo boundary without label.

Honesty application 16 — dispute resolution

Worked example 16 applying rocks table.

Dispute setup

Example 16: A vendor claims sub-micron SEM fidelity from spiderweb screenshot. Operator opens Chapter 12 table — procedural pixel detail, Metaphor label. Resolution: screenshot is Visual pedagogy, not lab.

Second dispute: jsonl shows SUSPICIOUS — colleague says GPU is compromised. Operator separates packet field local sockets from Vulkan dispatch — corroborate before permanent KILL.

Trust ordering

grep FieldRtxFieldAbs.hpp beats marketing PDF. run.log THERMO beats influencer stream. Default x86 die beats rumor about decorative engine.

Capstone equation layer

Landauer E_min is theory anchor — entropyThisFrame remains proxy. Shannon H on executable is NEXUS — not mixed into thermo boundary without label.

Honesty application 17 — dispute resolution

Worked example 17 applying rocks table.

Dispute setup

Example 17: A vendor claims sub-micron SEM fidelity from spiderweb screenshot. Operator opens Chapter 12 table — procedural pixel detail, Metaphor label. Resolution: screenshot is Visual pedagogy, not lab.

Second dispute: jsonl shows SUSPICIOUS — colleague says GPU is compromised. Operator separates packet field local sockets from Vulkan dispatch — corroborate before permanent KILL.

Trust ordering

grep FieldRtxFieldAbs.hpp beats marketing PDF. run.log THERMO beats influencer stream. Default x86 die beats rumor about decorative engine.

Capstone equation layer

Landauer E_min is theory anchor — entropyThisFrame remains proxy. Shannon H on executable is NEXUS — not mixed into thermo boundary without label.

Honesty application 18 — dispute resolution

Worked example 18 applying rocks table.

Dispute setup

Example 18: A vendor claims sub-micron SEM fidelity from spiderweb screenshot. Operator opens Chapter 12 table — procedural pixel detail, Metaphor label. Resolution: screenshot is Visual pedagogy, not lab.

Second dispute: jsonl shows SUSPICIOUS — colleague says GPU is compromised. Operator separates packet field local sockets from Vulkan dispatch — corroborate before permanent KILL.

Trust ordering

grep FieldRtxFieldAbs.hpp beats marketing PDF. run.log THERMO beats influencer stream. Default x86 die beats rumor about decorative engine.

Capstone equation layer

Landauer E_min is theory anchor — entropyThisFrame remains proxy. Shannon H on executable is NEXUS — not mixed into thermo boundary without label.

Honesty application 19 — dispute resolution

Worked example 19 applying rocks table.

Dispute setup

Example 19: A vendor claims sub-micron SEM fidelity from spiderweb screenshot. Operator opens Chapter 12 table — procedural pixel detail, Metaphor label. Resolution: screenshot is Visual pedagogy, not lab.

Second dispute: jsonl shows SUSPICIOUS — colleague says GPU is compromised. Operator separates packet field local sockets from Vulkan dispatch — corroborate before permanent KILL.

Trust ordering

grep FieldRtxFieldAbs.hpp beats marketing PDF. run.log THERMO beats influencer stream. Default x86 die beats rumor about decorative engine.

Capstone equation layer

Landauer E_min is theory anchor — entropyThisFrame remains proxy. Shannon H on executable is NEXUS — not mixed into thermo boundary without label.

Honesty application 20 — dispute resolution

Worked example 20 applying rocks table.

Dispute setup

Example 20: A vendor claims sub-micron SEM fidelity from spiderweb screenshot. Operator opens Chapter 12 table — procedural pixel detail, Metaphor label. Resolution: screenshot is Visual pedagogy, not lab.

Second dispute: jsonl shows SUSPICIOUS — colleague says GPU is compromised. Operator separates packet field local sockets from Vulkan dispatch — corroborate before permanent KILL.

Trust ordering

grep FieldRtxFieldAbs.hpp beats marketing PDF. run.log THERMO beats influencer stream. Default x86 die beats rumor about decorative engine.

Capstone equation layer

Landauer E_min is theory anchor — entropyThisFrame remains proxy. Shannon H on executable is NEXUS — not mixed into thermo boundary without label.

Honesty application 21 — dispute resolution

Worked example 21 applying rocks table.

Dispute setup

Example 21: A vendor claims sub-micron SEM fidelity from spiderweb screenshot. Operator opens Chapter 12 table — procedural pixel detail, Metaphor label. Resolution: screenshot is Visual pedagogy, not lab.

Second dispute: jsonl shows SUSPICIOUS — colleague says GPU is compromised. Operator separates packet field local sockets from Vulkan dispatch — corroborate before permanent KILL.

Trust ordering

grep FieldRtxFieldAbs.hpp beats marketing PDF. run.log THERMO beats influencer stream. Default x86 die beats rumor about decorative engine.

Capstone equation layer

Landauer E_min is theory anchor — entropyThisFrame remains proxy. Shannon H on executable is NEXUS — not mixed into thermo boundary without label.

Honesty application 22 — dispute resolution

Worked example 22 applying rocks table.

Dispute setup

Example 22: A vendor claims sub-micron SEM fidelity from spiderweb screenshot. Operator opens Chapter 12 table — procedural pixel detail, Metaphor label. Resolution: screenshot is Visual pedagogy, not lab.

Second dispute: jsonl shows SUSPICIOUS — colleague says GPU is compromised. Operator separates packet field local sockets from Vulkan dispatch — corroborate before permanent KILL.

Trust ordering

grep FieldRtxFieldAbs.hpp beats marketing PDF. run.log THERMO beats influencer stream. Default x86 die beats rumor about decorative engine.

Capstone equation layer

Landauer E_min is theory anchor — entropyThisFrame remains proxy. Shannon H on executable is NEXUS — not mixed into thermo boundary without label.

Honesty application 23 — dispute resolution

Worked example 23 applying rocks table.

Dispute setup

Example 23: A vendor claims sub-micron SEM fidelity from spiderweb screenshot. Operator opens Chapter 12 table — procedural pixel detail, Metaphor label. Resolution: screenshot is Visual pedagogy, not lab.

Second dispute: jsonl shows SUSPICIOUS — colleague says GPU is compromised. Operator separates packet field local sockets from Vulkan dispatch — corroborate before permanent KILL.

Trust ordering

grep FieldRtxFieldAbs.hpp beats marketing PDF. run.log THERMO beats influencer stream. Default x86 die beats rumor about decorative engine.

Capstone equation layer

Landauer E_min is theory anchor — entropyThisFrame remains proxy. Shannon H on executable is NEXUS — not mixed into thermo boundary without label.

Honesty application 24 — dispute resolution

Worked example 24 applying rocks table.

Dispute setup

Example 24: A vendor claims sub-micron SEM fidelity from spiderweb screenshot. Operator opens Chapter 12 table — procedural pixel detail, Metaphor label. Resolution: screenshot is Visual pedagogy, not lab.

Second dispute: jsonl shows SUSPICIOUS — colleague says GPU is compromised. Operator separates packet field local sockets from Vulkan dispatch — corroborate before permanent KILL.

Trust ordering

grep FieldRtxFieldAbs.hpp beats marketing PDF. run.log THERMO beats influencer stream. Default x86 die beats rumor about decorative engine.

Capstone equation layer

Landauer E_min is theory anchor — entropyThisFrame remains proxy. Shannon H on executable is NEXUS — not mixed into thermo boundary without label.

Honesty application 25 — dispute resolution

Worked example 25 applying rocks table.

Dispute setup

Example 25: A vendor claims sub-micron SEM fidelity from spiderweb screenshot. Operator opens Chapter 12 table — procedural pixel detail, Metaphor label. Resolution: screenshot is Visual pedagogy, not lab.

Second dispute: jsonl shows SUSPICIOUS — colleague says GPU is compromised. Operator separates packet field local sockets from Vulkan dispatch — corroborate before permanent KILL.

Trust ordering

grep FieldRtxFieldAbs.hpp beats marketing PDF. run.log THERMO beats influencer stream. Default x86 die beats rumor about decorative engine.

Capstone equation layer

Landauer E_min is theory anchor — entropyThisFrame remains proxy. Shannon H on executable is NEXUS — not mixed into thermo boundary without label.

Honesty application 26 — dispute resolution

Worked example 26 applying rocks table.

Dispute setup

Example 26: A vendor claims sub-micron SEM fidelity from spiderweb screenshot. Operator opens Chapter 12 table — procedural pixel detail, Metaphor label. Resolution: screenshot is Visual pedagogy, not lab.

Second dispute: jsonl shows SUSPICIOUS — colleague says GPU is compromised. Operator separates packet field local sockets from Vulkan dispatch — corroborate before permanent KILL.

Trust ordering

grep FieldRtxFieldAbs.hpp beats marketing PDF. run.log THERMO beats influencer stream. Default x86 die beats rumor about decorative engine.

Capstone equation layer

Landauer E_min is theory anchor — entropyThisFrame remains proxy. Shannon H on executable is NEXUS — not mixed into thermo boundary without label.

Honesty application 27 — dispute resolution

Worked example 27 applying rocks table.

Dispute setup

Example 27: A vendor claims sub-micron SEM fidelity from spiderweb screenshot. Operator opens Chapter 12 table — procedural pixel detail, Metaphor label. Resolution: screenshot is Visual pedagogy, not lab.

Second dispute: jsonl shows SUSPICIOUS — colleague says GPU is compromised. Operator separates packet field local sockets from Vulkan dispatch — corroborate before permanent KILL.

Trust ordering

grep FieldRtxFieldAbs.hpp beats marketing PDF. run.log THERMO beats influencer stream. Default x86 die beats rumor about decorative engine.

Capstone equation layer

Landauer E_min is theory anchor — entropyThisFrame remains proxy. Shannon H on executable is NEXUS — not mixed into thermo boundary without label.

Honesty application 28 — dispute resolution

Worked example 28 applying rocks table.

Dispute setup

Example 28: A vendor claims sub-micron SEM fidelity from spiderweb screenshot. Operator opens Chapter 12 table — procedural pixel detail, Metaphor label. Resolution: screenshot is Visual pedagogy, not lab.

Second dispute: jsonl shows SUSPICIOUS — colleague says GPU is compromised. Operator separates packet field local sockets from Vulkan dispatch — corroborate before permanent KILL.

Trust ordering

grep FieldRtxFieldAbs.hpp beats marketing PDF. run.log THERMO beats influencer stream. Default x86 die beats rumor about decorative engine.

Capstone equation layer

Landauer E_min is theory anchor — entropyThisFrame remains proxy. Shannon H on executable is NEXUS — not mixed into thermo boundary without label.

Honesty application 29 — dispute resolution

Worked example 29 applying rocks table.

Dispute setup

Example 29: A vendor claims sub-micron SEM fidelity from spiderweb screenshot. Operator opens Chapter 12 table — procedural pixel detail, Metaphor label. Resolution: screenshot is Visual pedagogy, not lab.

Second dispute: jsonl shows SUSPICIOUS — colleague says GPU is compromised. Operator separates packet field local sockets from Vulkan dispatch — corroborate before permanent KILL.

Trust ordering

grep FieldRtxFieldAbs.hpp beats marketing PDF. run.log THERMO beats influencer stream. Default x86 die beats rumor about decorative engine.

Capstone equation layer

Landauer E_min is theory anchor — entropyThisFrame remains proxy. Shannon H on executable is NEXUS — not mixed into thermo boundary without label.

Honesty application 30 — dispute resolution

Worked example 30 applying rocks table.

Dispute setup

Example 30: A vendor claims sub-micron SEM fidelity from spiderweb screenshot. Operator opens Chapter 12 table — procedural pixel detail, Metaphor label. Resolution: screenshot is Visual pedagogy, not lab.

Second dispute: jsonl shows SUSPICIOUS — colleague says GPU is compromised. Operator separates packet field local sockets from Vulkan dispatch — corroborate before permanent KILL.

Trust ordering

grep FieldRtxFieldAbs.hpp beats marketing PDF. run.log THERMO beats influencer stream. Default x86 die beats rumor about decorative engine.

Capstone equation layer

Landauer E_min is theory anchor — entropyThisFrame remains proxy. Shannon H on executable is NEXUS — not mixed into thermo boundary without label.

Honesty application 31 — dispute resolution

Worked example 31 applying rocks table.

Dispute setup

Example 31: A vendor claims sub-micron SEM fidelity from spiderweb screenshot. Operator opens Chapter 12 table — procedural pixel detail, Metaphor label. Resolution: screenshot is Visual pedagogy, not lab.

Second dispute: jsonl shows SUSPICIOUS — colleague says GPU is compromised. Operator separates packet field local sockets from Vulkan dispatch — corroborate before permanent KILL.

Trust ordering

grep FieldRtxFieldAbs.hpp beats marketing PDF. run.log THERMO beats influencer stream. Default x86 die beats rumor about decorative engine.

Capstone equation layer

Landauer E_min is theory anchor — entropyThisFrame remains proxy. Shannon H on executable is NEXUS — not mixed into thermo boundary without label.

Honesty application 32 — dispute resolution

Worked example 32 applying rocks table.

Dispute setup

Example 32: A vendor claims sub-micron SEM fidelity from spiderweb screenshot. Operator opens Chapter 12 table — procedural pixel detail, Metaphor label. Resolution: screenshot is Visual pedagogy, not lab.

Second dispute: jsonl shows SUSPICIOUS — colleague says GPU is compromised. Operator separates packet field local sockets from Vulkan dispatch — corroborate before permanent KILL.

Trust ordering

grep FieldRtxFieldAbs.hpp beats marketing PDF. run.log THERMO beats influencer stream. Default x86 die beats rumor about decorative engine.

Capstone equation layer

Landauer E_min is theory anchor — entropyThisFrame remains proxy. Shannon H on executable is NEXUS — not mixed into thermo boundary without label.

Honesty application 33 — dispute resolution

Worked example 33 applying rocks table.

Dispute setup

Example 33: A vendor claims sub-micron SEM fidelity from spiderweb screenshot. Operator opens Chapter 12 table — procedural pixel detail, Metaphor label. Resolution: screenshot is Visual pedagogy, not lab.

Second dispute: jsonl shows SUSPICIOUS — colleague says GPU is compromised. Operator separates packet field local sockets from Vulkan dispatch — corroborate before permanent KILL.

Trust ordering

grep FieldRtxFieldAbs.hpp beats marketing PDF. run.log THERMO beats influencer stream. Default x86 die beats rumor about decorative engine.

Capstone equation layer

Landauer E_min is theory anchor — entropyThisFrame remains proxy. Shannon H on executable is NEXUS — not mixed into thermo boundary without label.

Honesty application 34 — dispute resolution

Worked example 34 applying rocks table.

Dispute setup

Example 34: A vendor claims sub-micron SEM fidelity from spiderweb screenshot. Operator opens Chapter 12 table — procedural pixel detail, Metaphor label. Resolution: screenshot is Visual pedagogy, not lab.

Second dispute: jsonl shows SUSPICIOUS — colleague says GPU is compromised. Operator separates packet field local sockets from Vulkan dispatch — corroborate before permanent KILL.

Trust ordering

grep FieldRtxFieldAbs.hpp beats marketing PDF. run.log THERMO beats influencer stream. Default x86 die beats rumor about decorative engine.

Capstone equation layer

Landauer E_min is theory anchor — entropyThisFrame remains proxy. Shannon H on executable is NEXUS — not mixed into thermo boundary without label.

Honesty application 35 — dispute resolution

Worked example 35 applying rocks table.

Dispute setup

Example 35: A vendor claims sub-micron SEM fidelity from spiderweb screenshot. Operator opens Chapter 12 table — procedural pixel detail, Metaphor label. Resolution: screenshot is Visual pedagogy, not lab.

Second dispute: jsonl shows SUSPICIOUS — colleague says GPU is compromised. Operator separates packet field local sockets from Vulkan dispatch — corroborate before permanent KILL.

Trust ordering

grep FieldRtxFieldAbs.hpp beats marketing PDF. run.log THERMO beats influencer stream. Default x86 die beats rumor about decorative engine.

Capstone equation layer

Landauer E_min is theory anchor — entropyThisFrame remains proxy. Shannon H on executable is NEXUS — not mixed into thermo boundary without label.

Honesty application 36 — dispute resolution

Worked example 36 applying rocks table.

Dispute setup

Example 36: A vendor claims sub-micron SEM fidelity from spiderweb screenshot. Operator opens Chapter 12 table — procedural pixel detail, Metaphor label. Resolution: screenshot is Visual pedagogy, not lab.

Second dispute: jsonl shows SUSPICIOUS — colleague says GPU is compromised. Operator separates packet field local sockets from Vulkan dispatch — corroborate before permanent KILL.

Trust ordering

grep FieldRtxFieldAbs.hpp beats marketing PDF. run.log THERMO beats influencer stream. Default x86 die beats rumor about decorative engine.

Capstone equation layer

Landauer E_min is theory anchor — entropyThisFrame remains proxy. Shannon H on executable is NEXUS — not mixed into thermo boundary without label.

Evidence anchor — grep and sources

Major claims in this chapter anchored for reproducibility. Implemented = grep today; Metaphor
= intuition; Philosophy = discipline.

Claim	Statement	Label	Evidence
Honesty table	Claim vs operator reality	Implemented	This chapter – bookmark
Trust order	Headers > stderr > art	Philosophy	Dispute resolution
Capstone equations	Layer-tagged math	Metaphor	Theory vs proxy vs file H
Product boundaries	Four licenses	Implemented	AMOURANTHRTX / NEXUS / Queen / Primer

See capstone table – CFL, Landauer, Shannon, entropy proxy

```
grep -r 'tag impl\|tag meta' docs/chapters/12-reality-theory.html
```

Source paths

- content/chapters/12.html
- scripts/master_rocks_table.py

Chapter summary — before you turn the page

Rocks visible. Labels mandatory. Return when sold cosmology, joule fantasy, or shader instrumentation.
This chapter is the honesty contract for the whole book.